

Steps to Upgrade SQL Server Cluster Instance from 2014 to 2016.

Lower: SQL Server 2014

Higher: SQL Server 2016

Node-1: Active

Node-2: Passive

- 1. Action Items**
- 2. Before Up-gradation**
- 3. Up-gradation Process**

Action Items:

Step-1: Upgrade the SQL Services in "**PASSIVE**" Node [Node-2].

Passive node rebooted.

Step-2: Post Up-gradation, failover the SQL Services from "**Active [Node-1]**" to "**Passive [Node-2]**" node.

After Failover:

Node-1: Passive

Node-2: Active

Step-3: Now upgrade the SQL Services on new "**Passive [Node-1]**" Node.

Step-4: Post up-gradation, failback the SQL Services from current new "**Active [Node-2]**" to "**Passive [Node-1]**".

After Failback:

Node-1: Active

Node-2: Passive

Step-5: Verify the SQL Version of newly upgraded SQL Cluster Instance using below TSQL.

Select @@Version

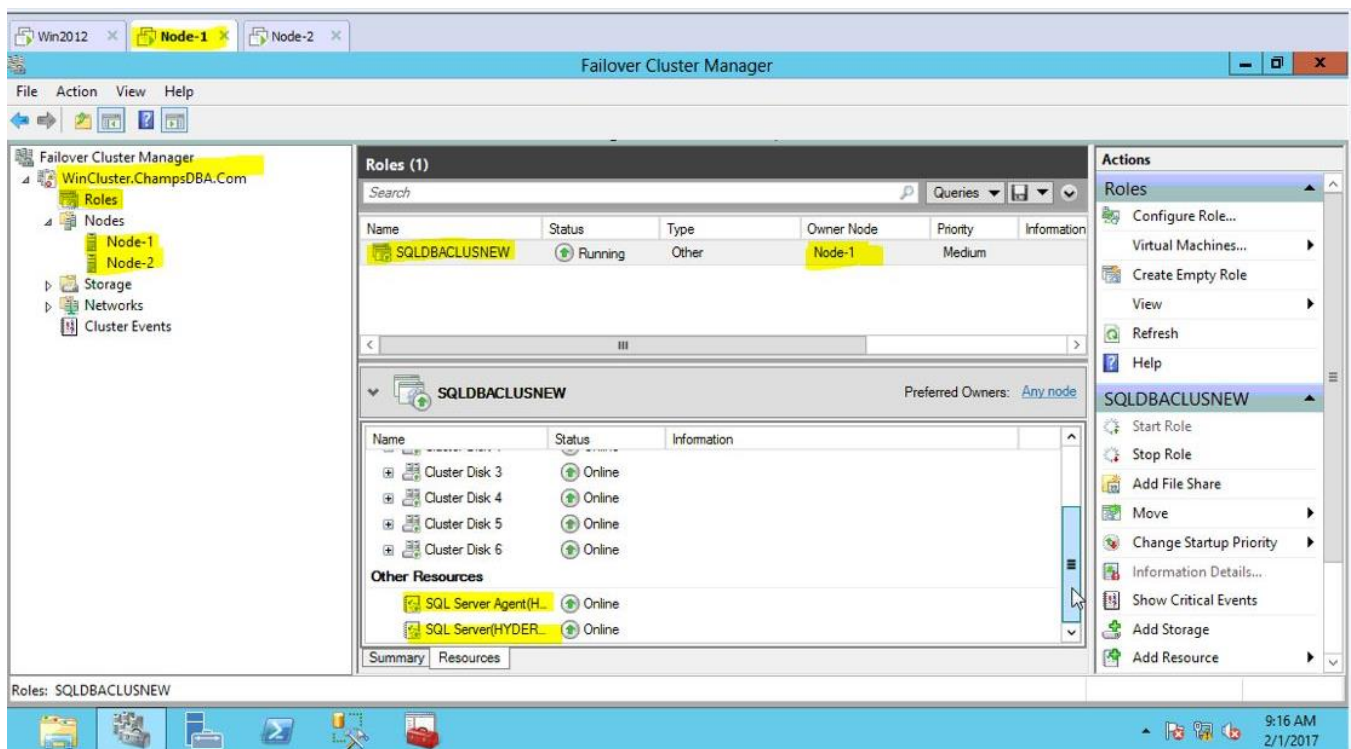
Before Up-gradation:

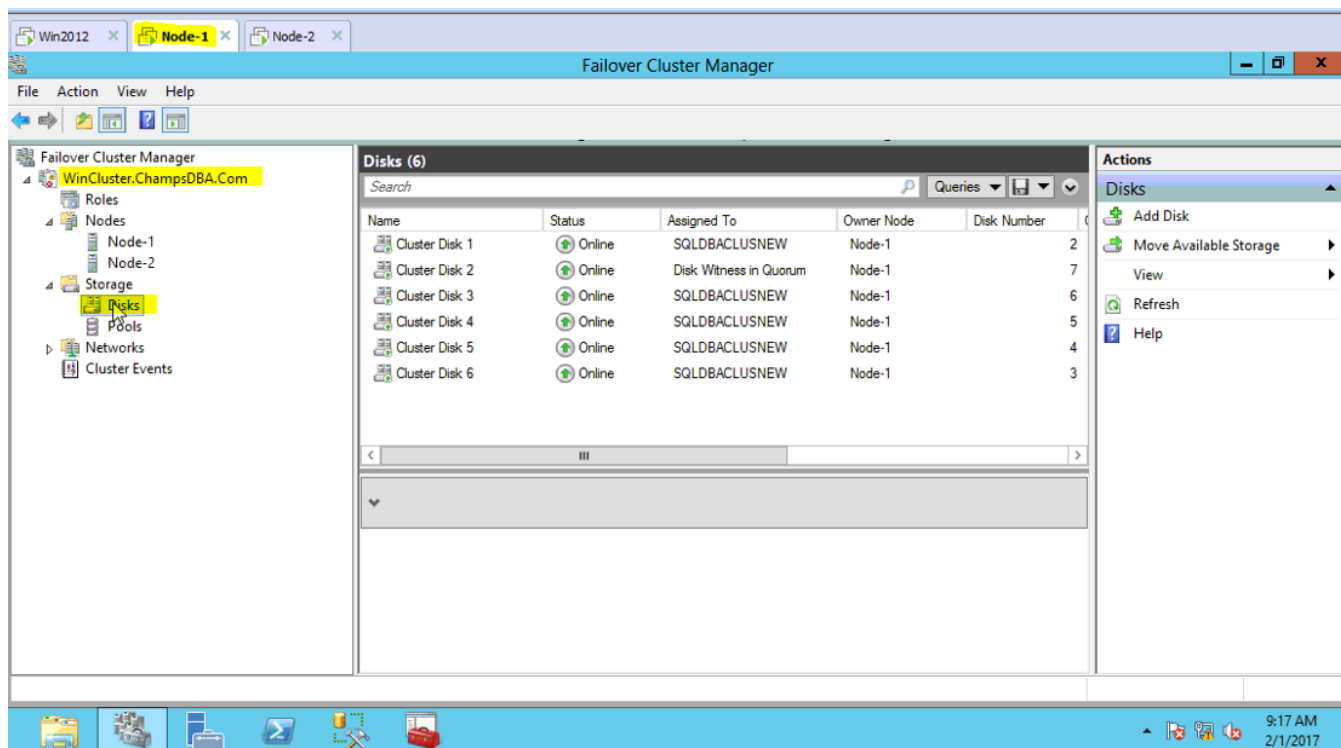
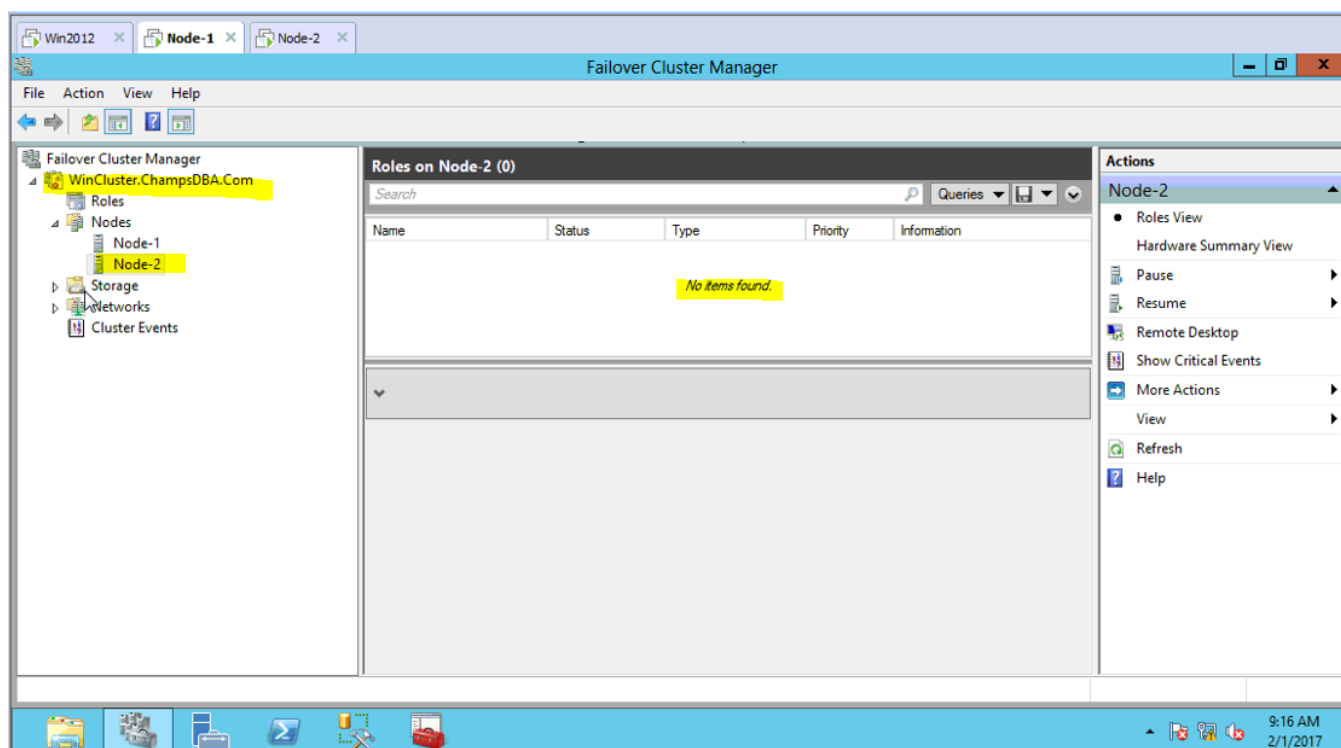
Active: Node-1

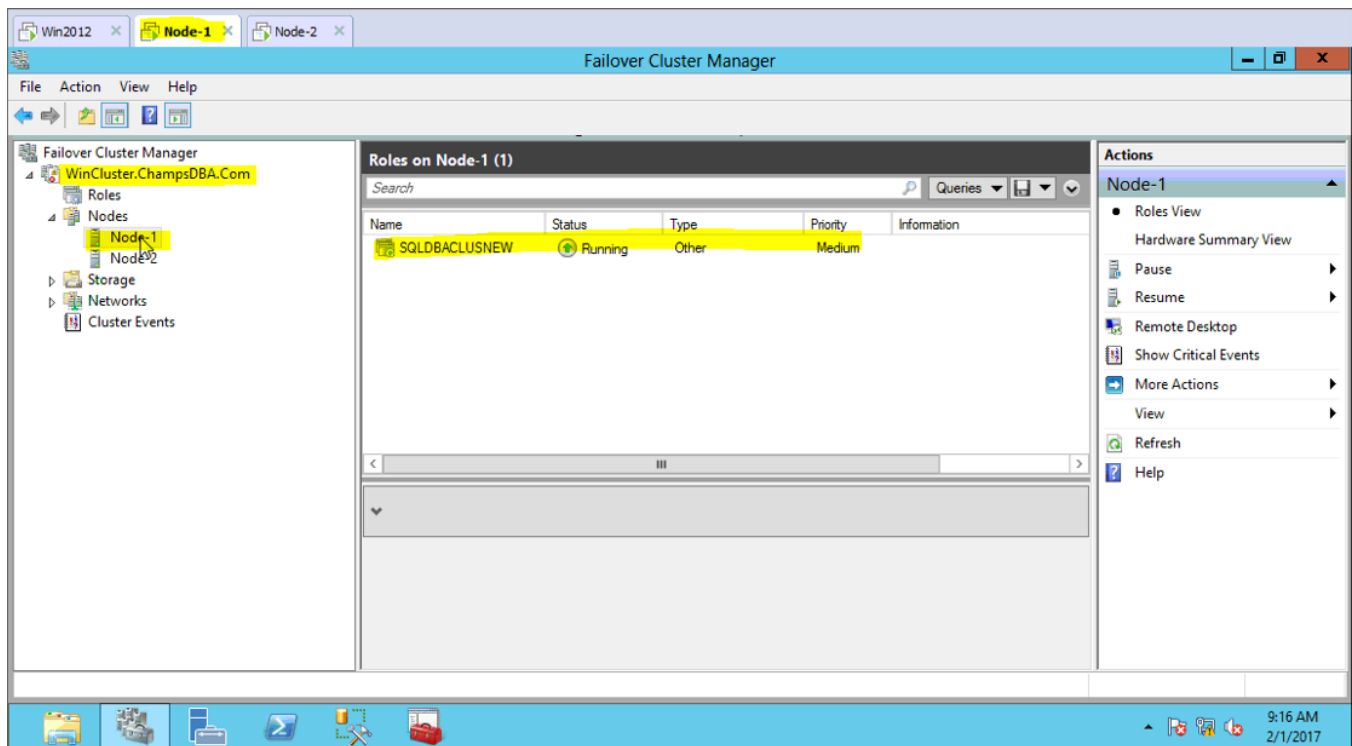
Passive: Node-2

Validation Before Up-gradation.

1. Go to any node and verify the services and its current owner details.







Node-1: Active

SQL Cluster Instance Name: SQLDBACLS\HYDERABAD

SQL Server Version Report:

Microsoft SQL Server 2014 - 12.0.2000.8 (X64)

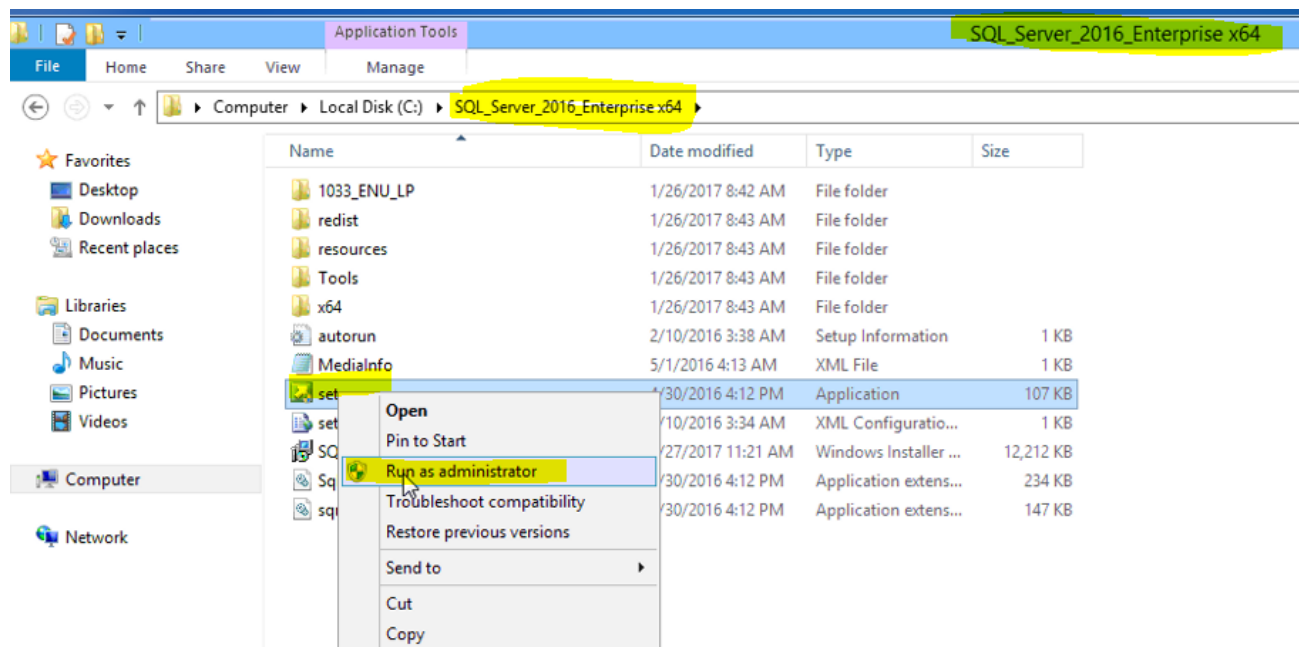
Feb 20 2014 20:04:26

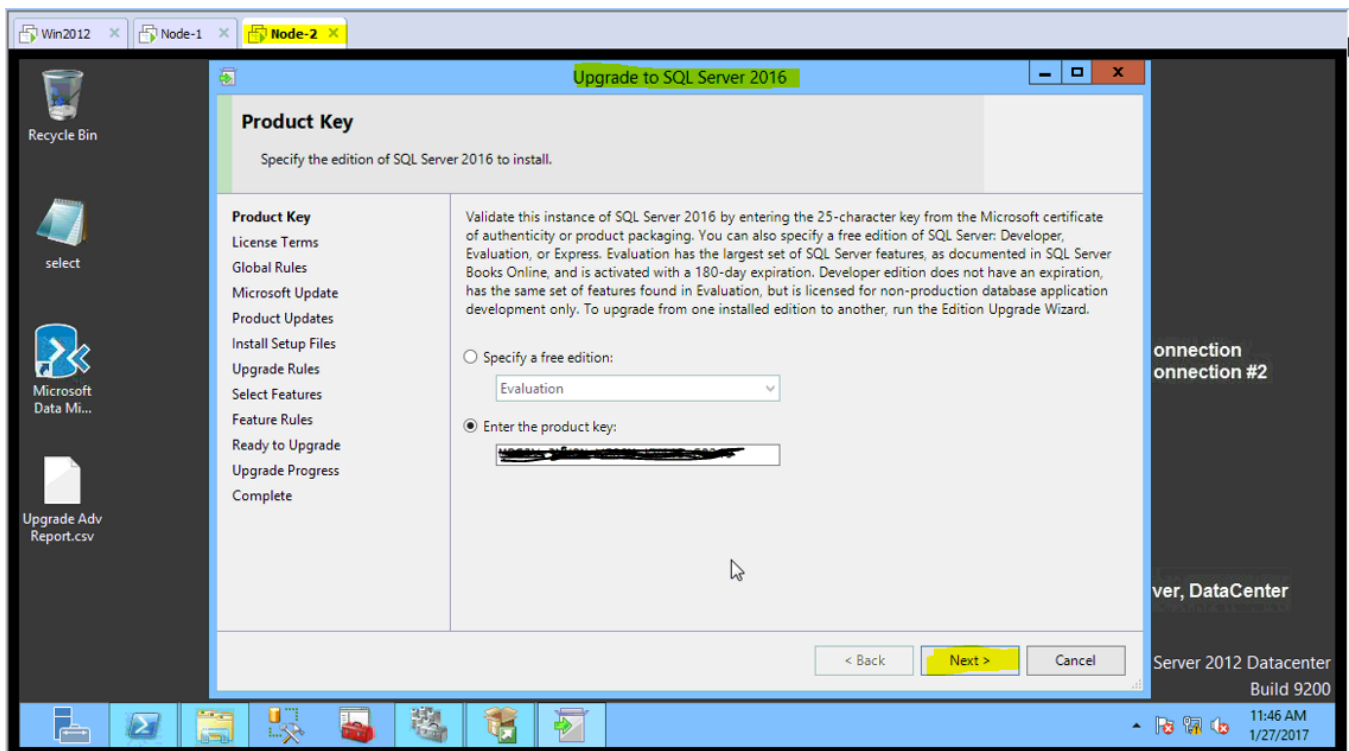
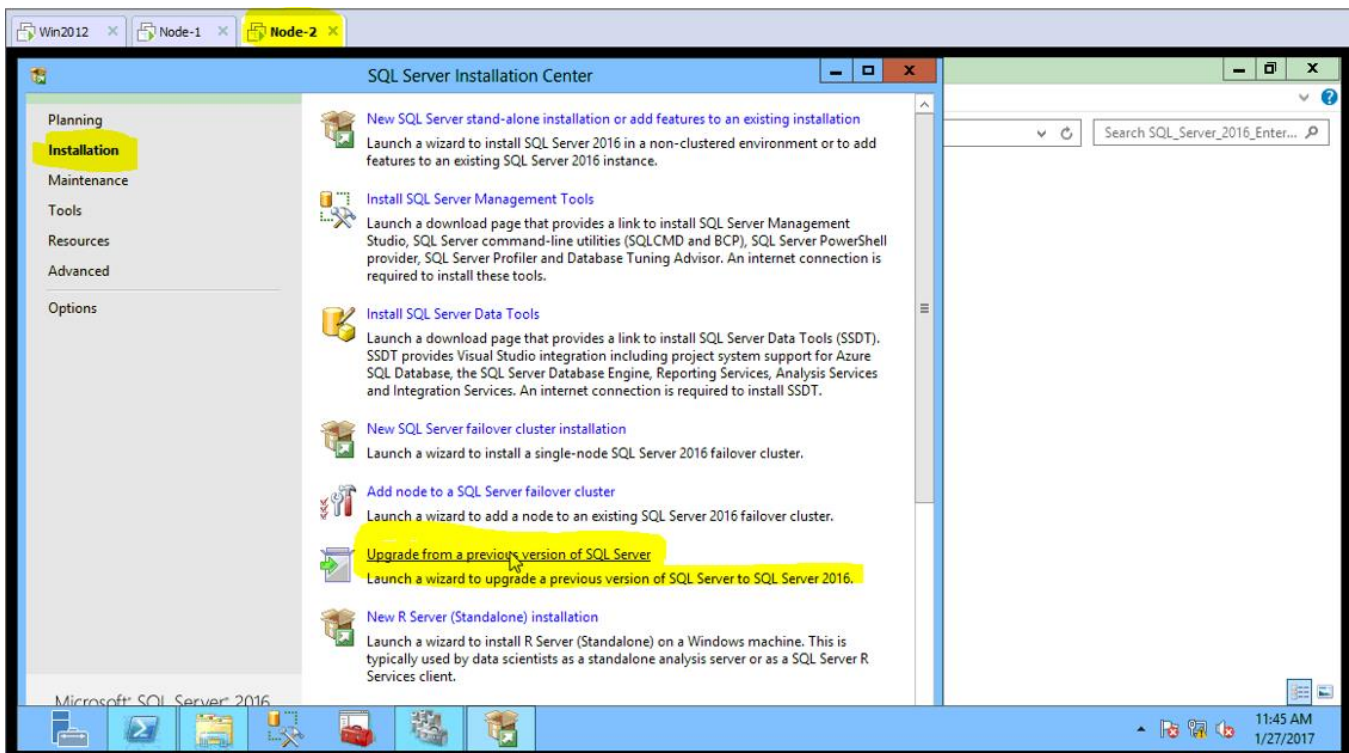
Copyright (c) Microsoft Corporation

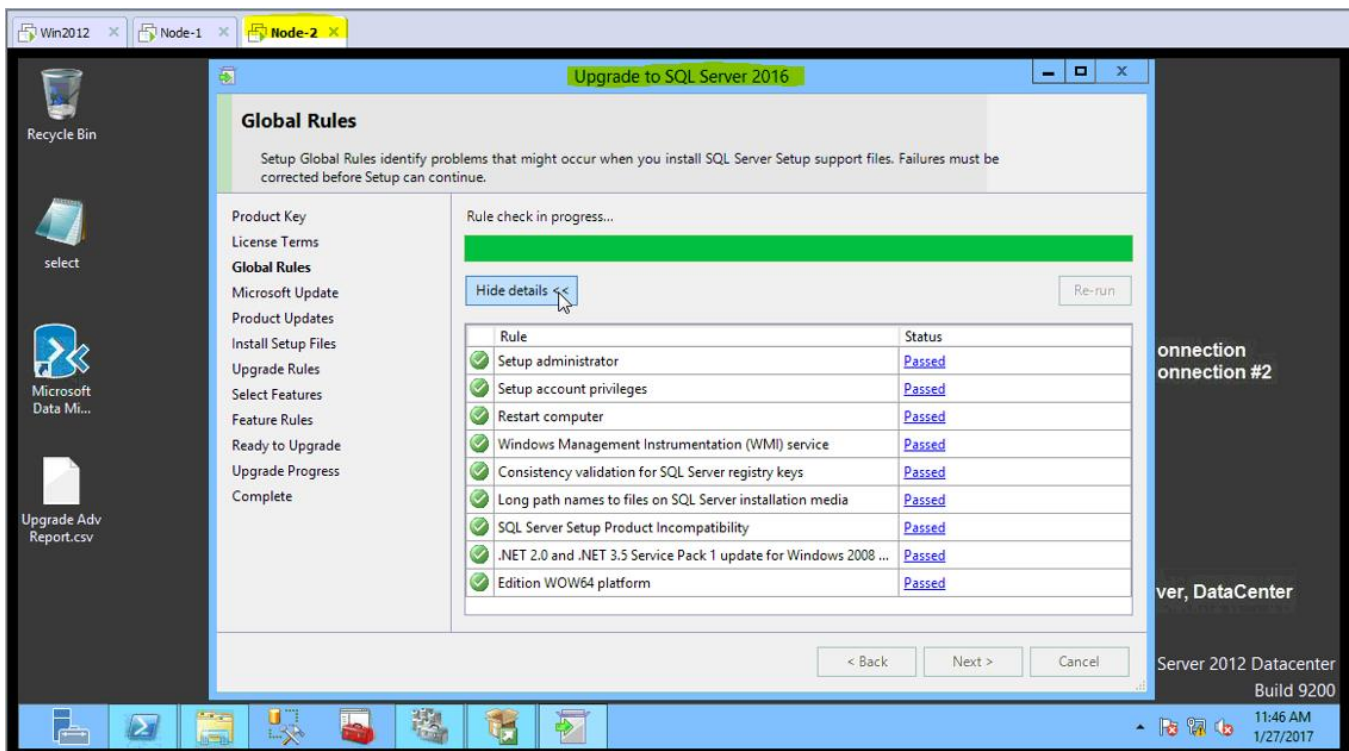
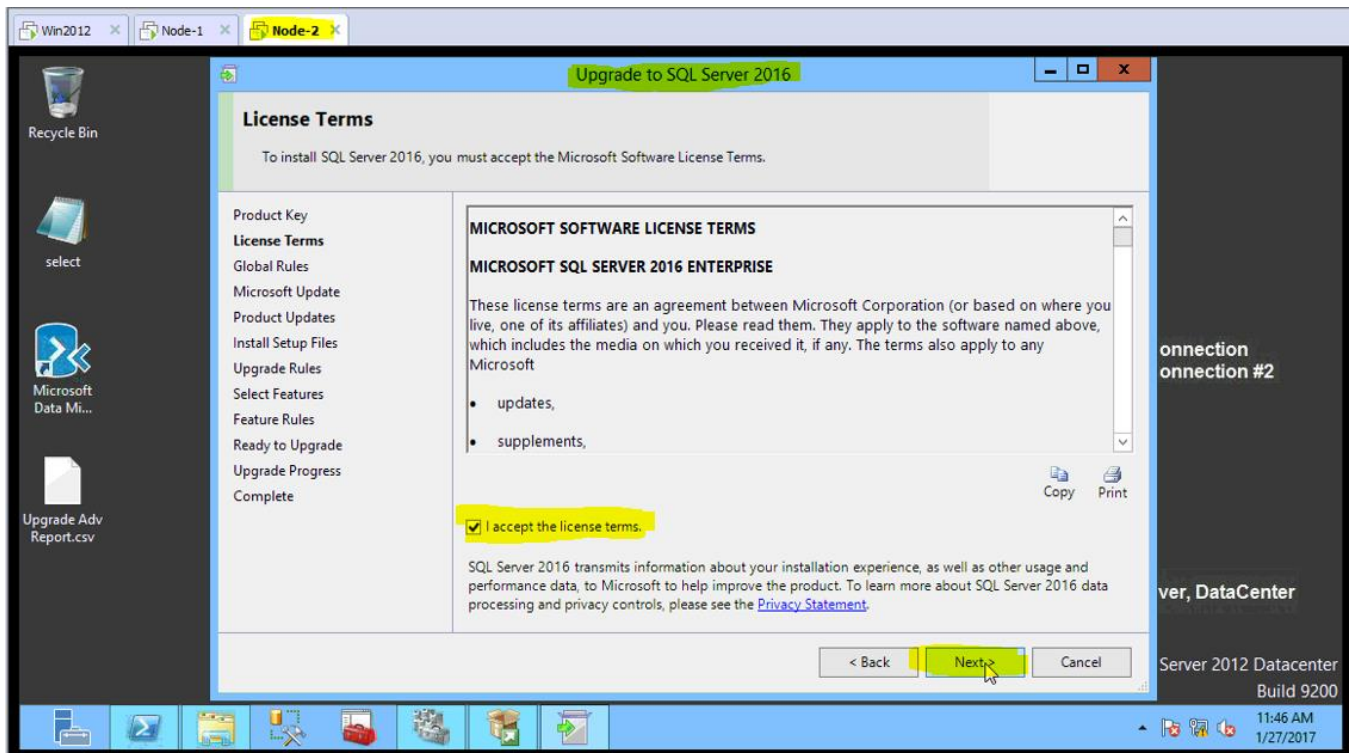
**Enterprise Edition: Core-based Licensing (64-bit) on Windows NT 6.2 <X64> (Build 9200:)
(Hypervisor)**

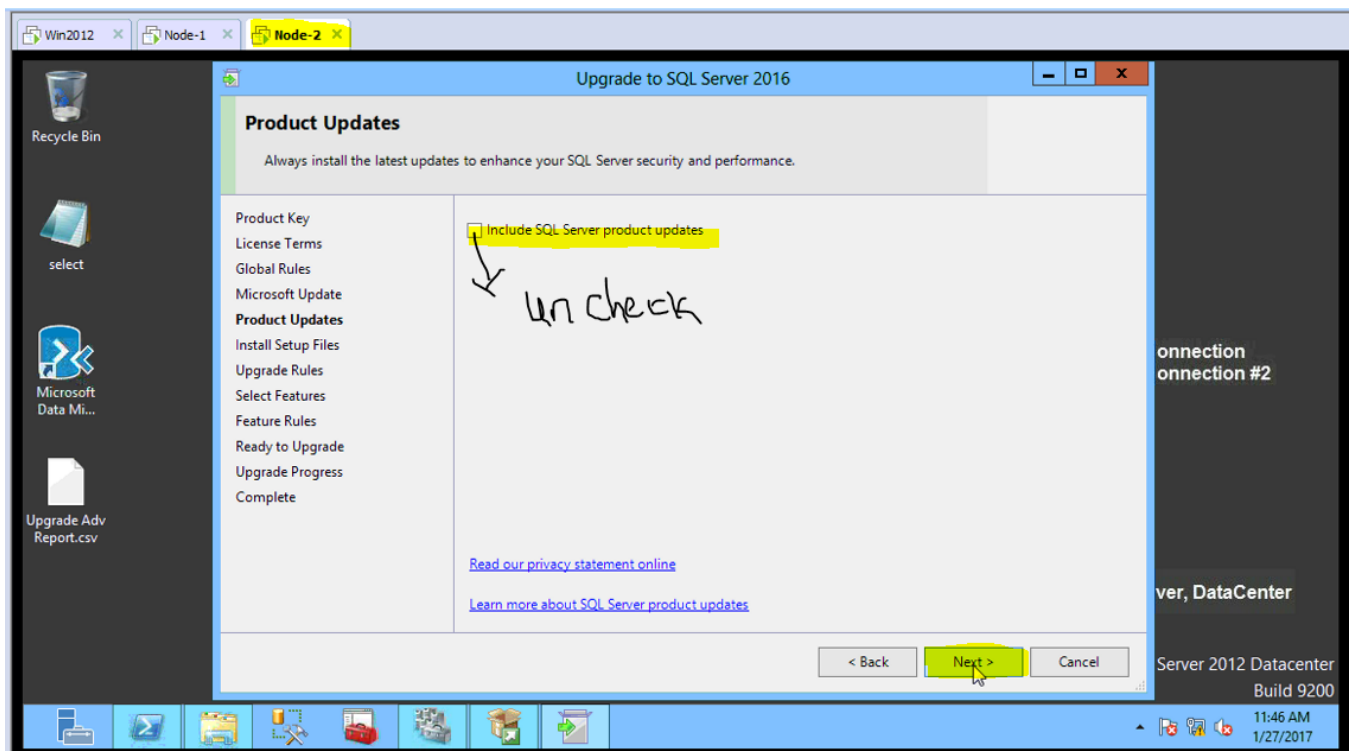
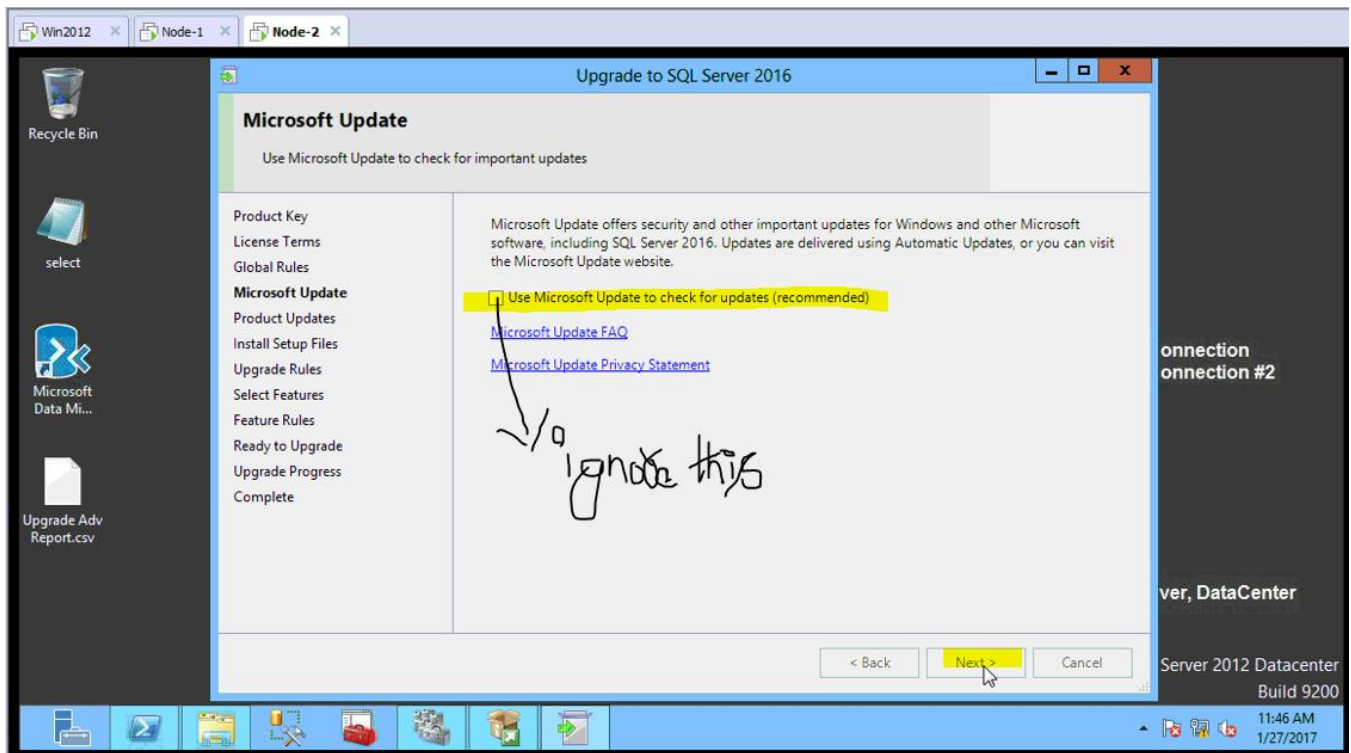
Up-Gradation Process:

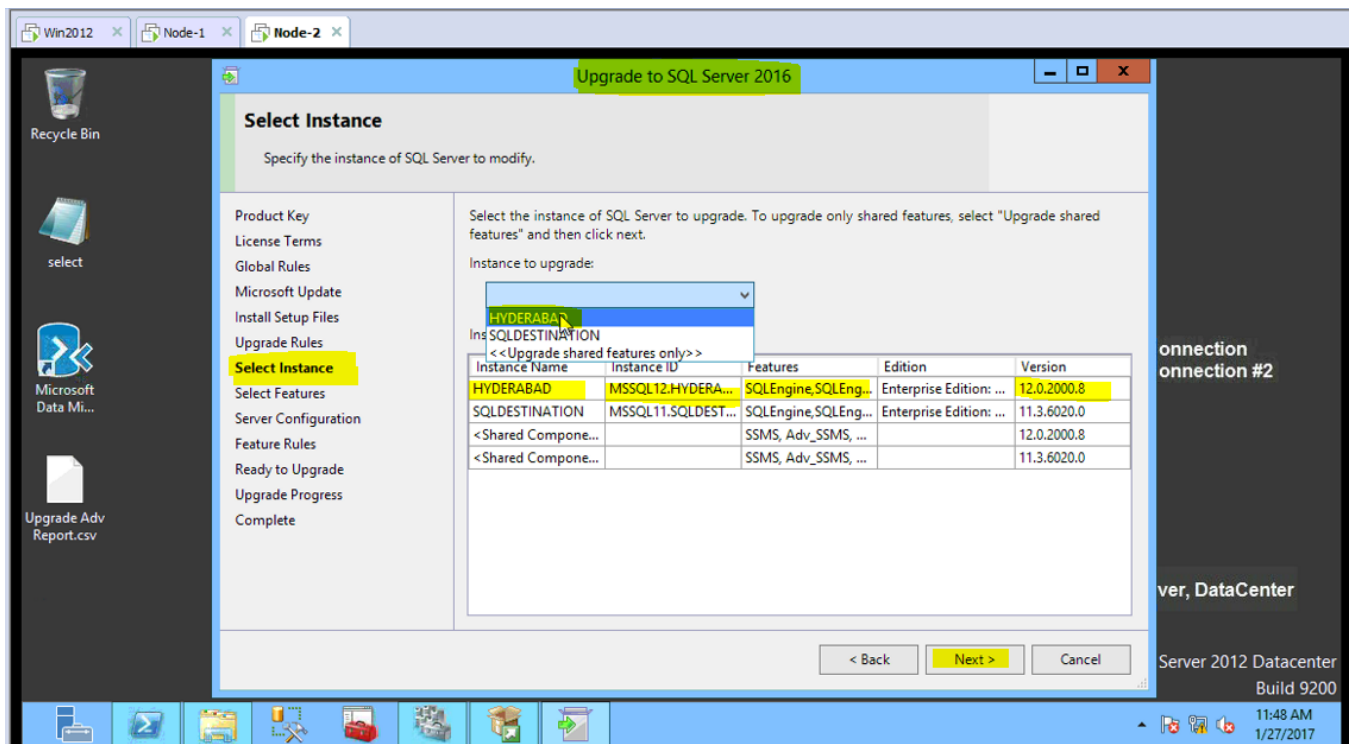
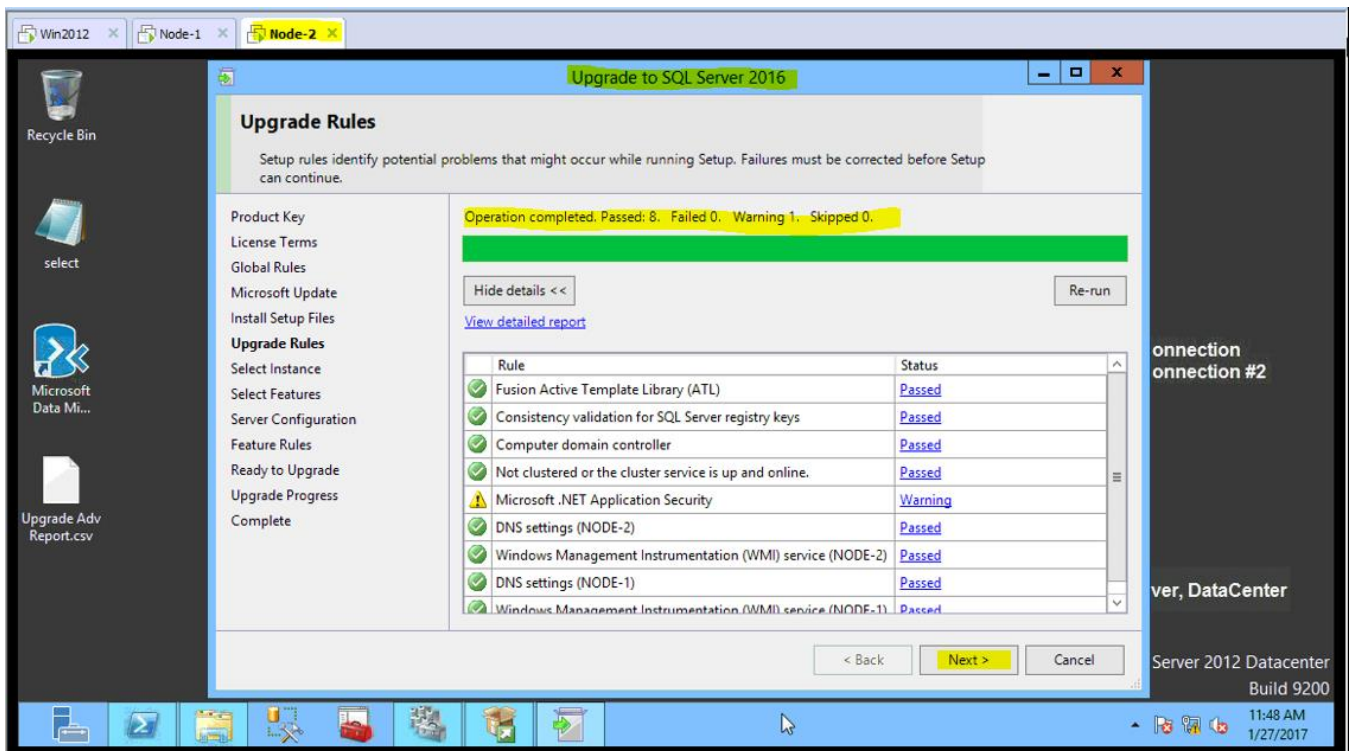
1. Take Remote Desktop Connection to "Passive" Node [Node-2].
2. Ensure that Node-2 should not own any services or disks.
3. Navigate to the Path where SQL Server 2016

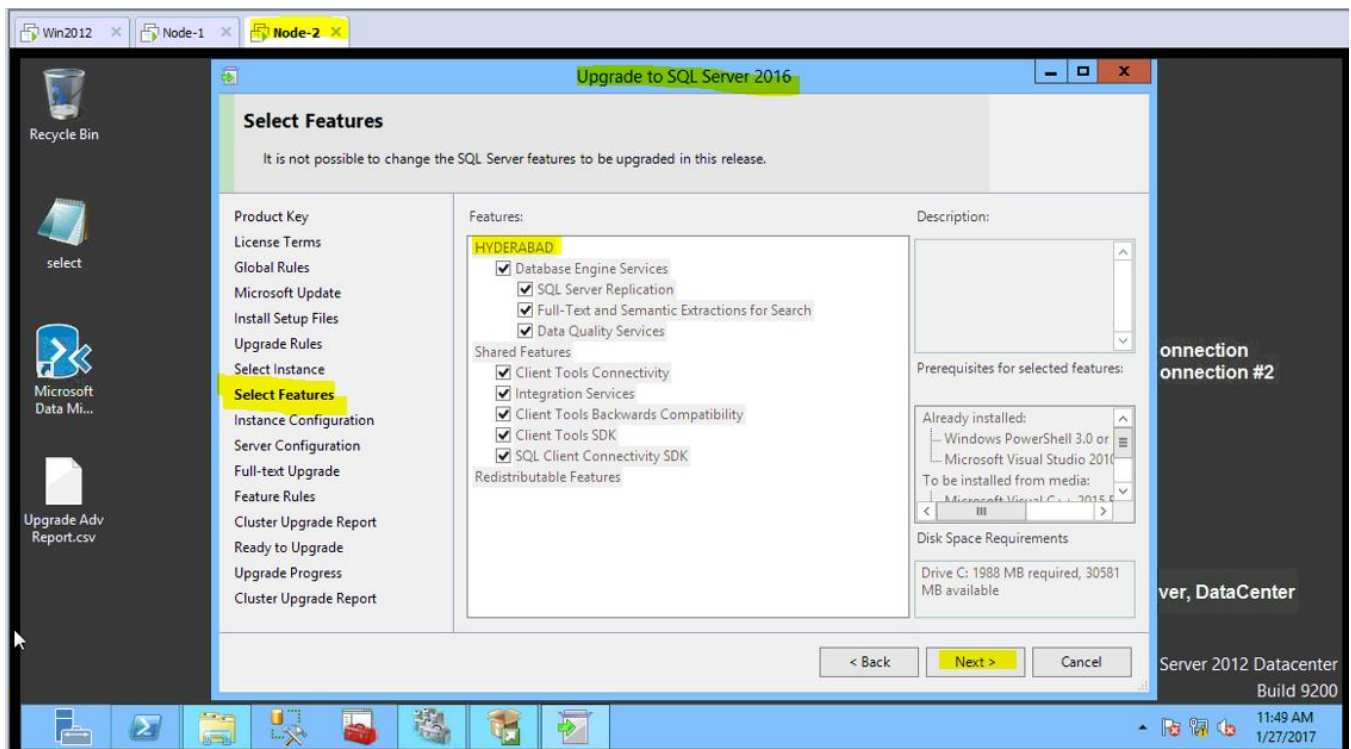
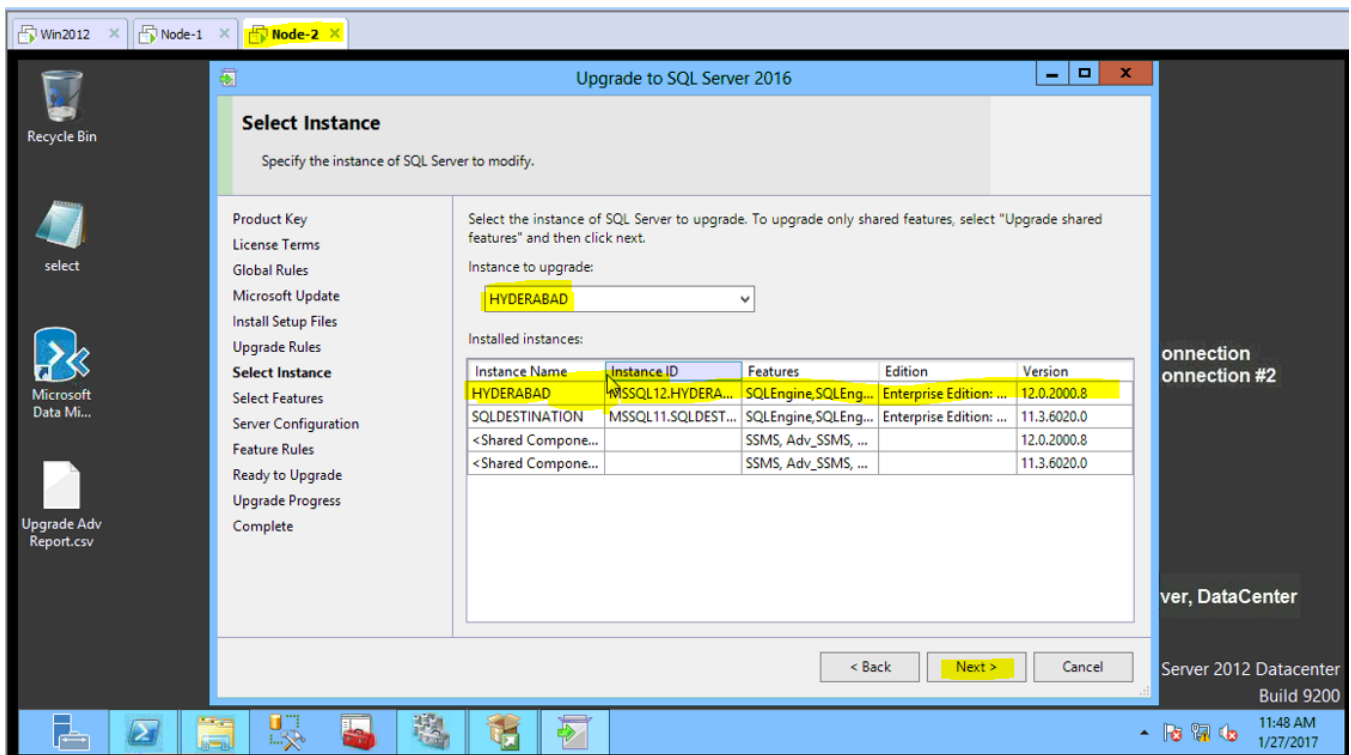












Win2012 x Node-1 x Node-2 x

Upgrade to SQL Server 2016

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Product Key
License Terms
Global Rules
Microsoft Update
Install Setup Files
Upgrade Rules
Select Instance
Select Features
Instance Configuration
Server Configuration
Full-text Upgrade
Feature Rules
Cluster Upgrade Report
Ready to Upgrade
Upgrade Progress
Cluster Upgrade Report
Complete

☐ Default instance
☒ Named instance: **HYDERABAD**

Instance ID: **HYDERABAD**

SQL Server directory: B:\Program Files\Microsoft SQL Server\MSSQL13.HYDERABAD

Installed instances:

Instance Name	Instance ID	Features	Edition	Version
HYDERABAD	MSSQL12.HYDERABAD	SQLEngine,SQLEngine\Replication,...	Enterprise Edition: Core-based Lice...	12.0.2000.8
SQLDESTINATION	MSSQL11.SQLDESTINATION	SQLEngine,SQLEngine\Replication,...	Enterprise Edition: Core-based Lice...	11.3.6020.0
<Shared Components>		SSMS, Adv_SSMS, Conn, BC, SDK, IS		12.0.2000.8
<Shared Components>		SSMS, Adv_SSMS, Conn, BC, SDK, IS		11.3.6020.0

< Back **Next >** Cancel

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Win2012 x Node-1 x Node-2 x

Upgrade to SQL Server 2016

Server Configuration

Specify the service accounts and collation configuration.

Product Key
License Terms
Global Rules
Microsoft Update
Install Setup Files
Upgrade Rules
Select Instance
Select Features
Instance Configuration
Server Configuration
Full-text Upgrade
Feature Rules
Cluster Upgrade Report
Ready to Upgrade
Upgrade Progress
Cluster Upgrade Report
Complete

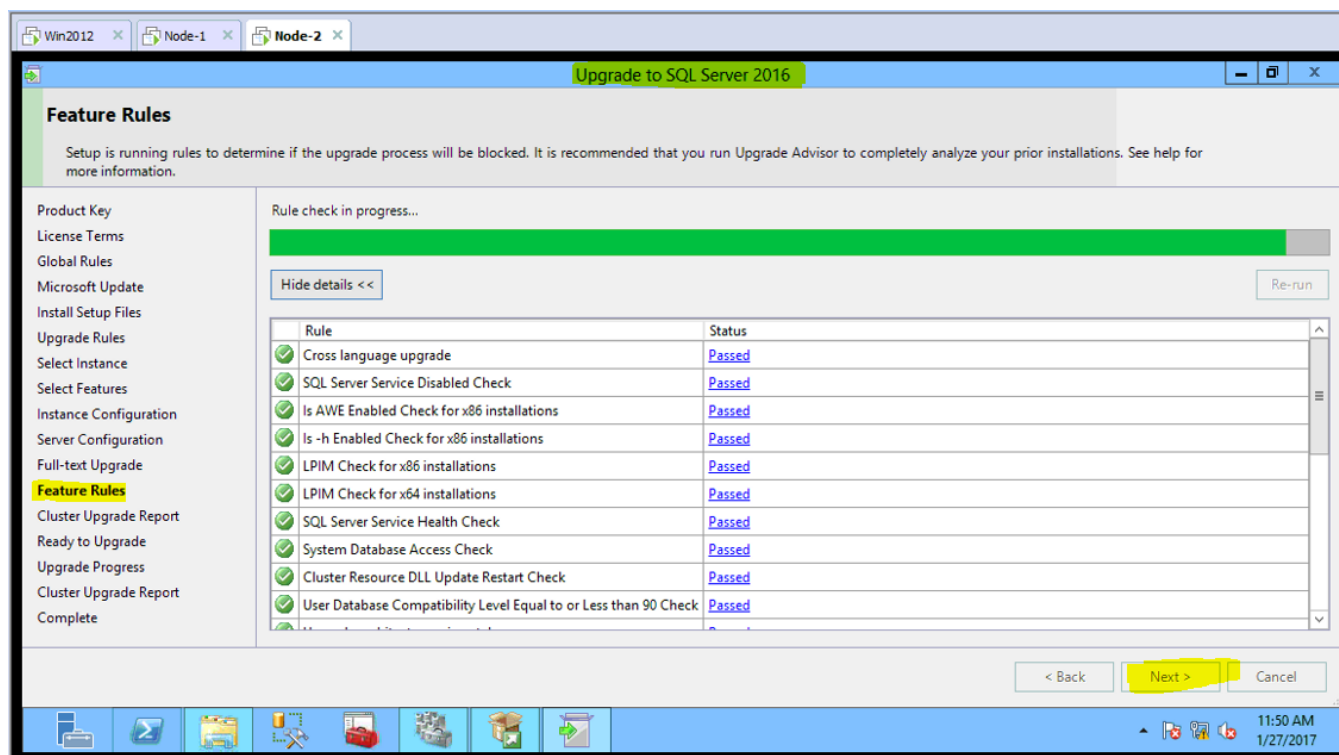
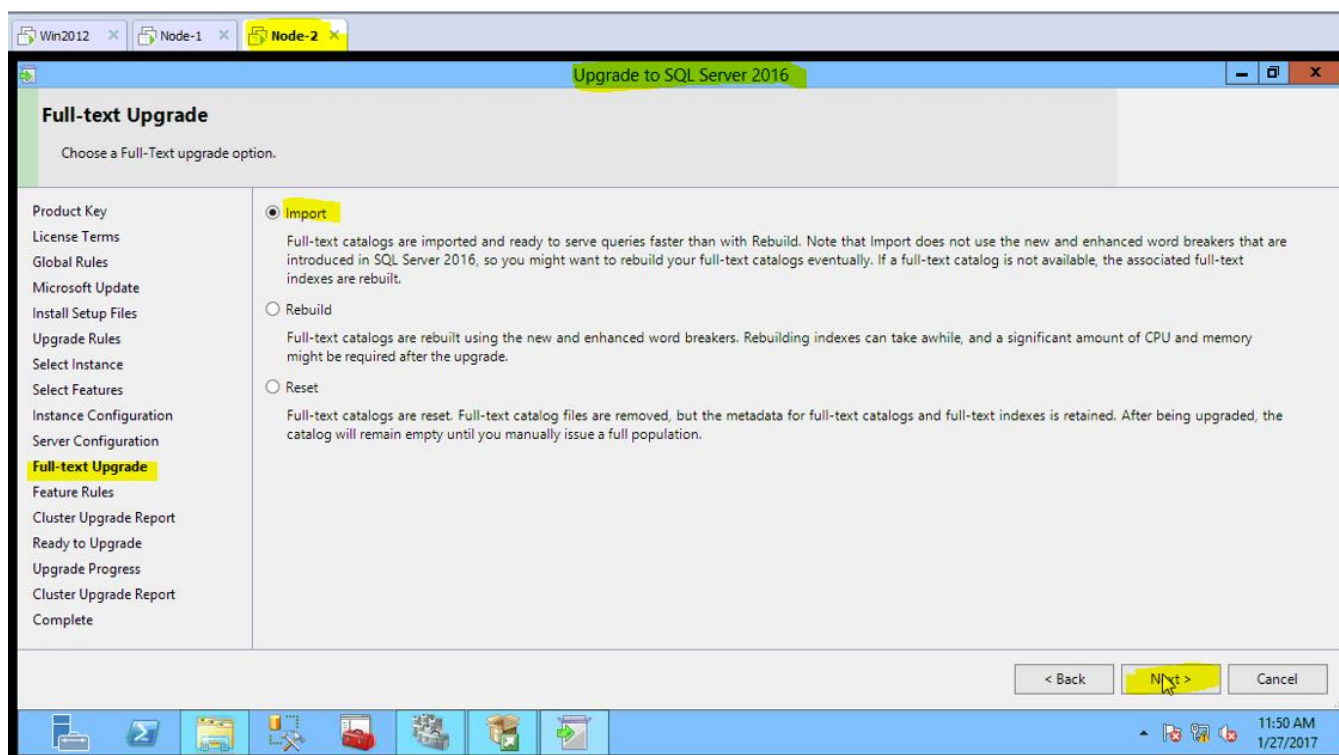
Service Accounts

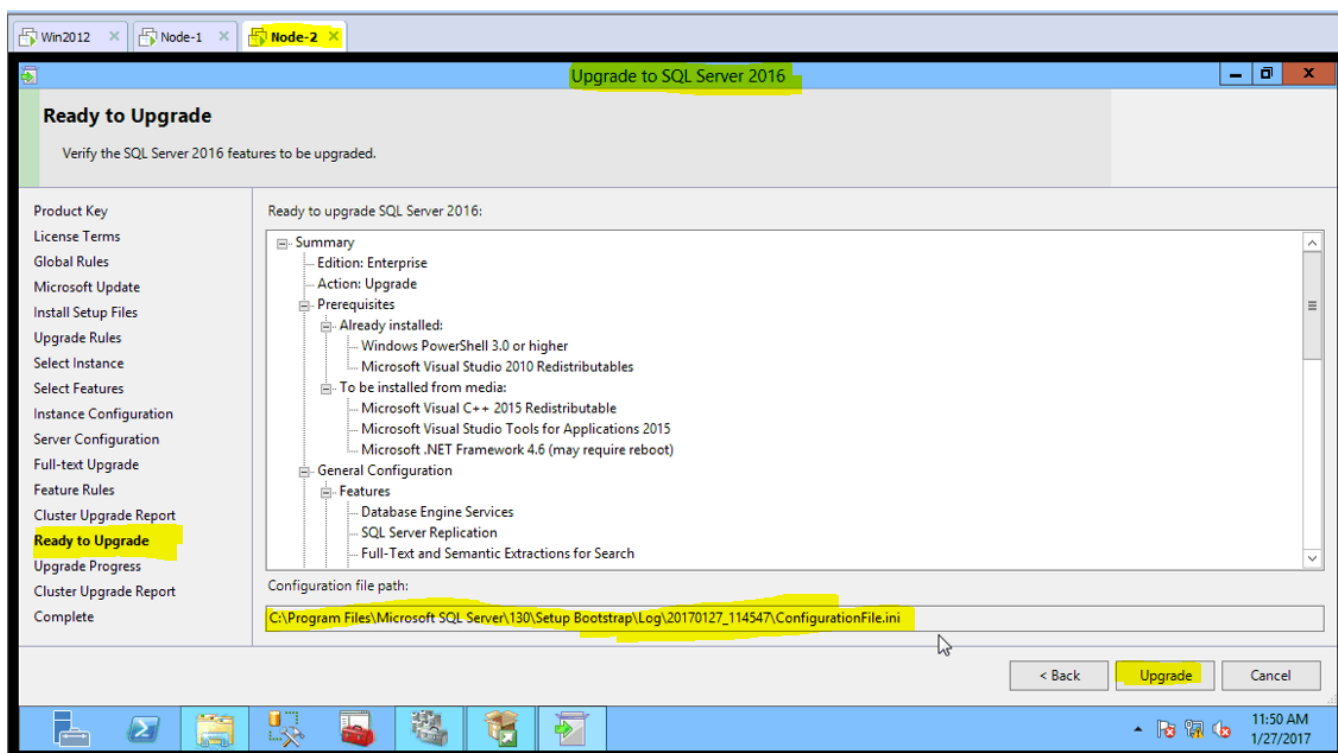
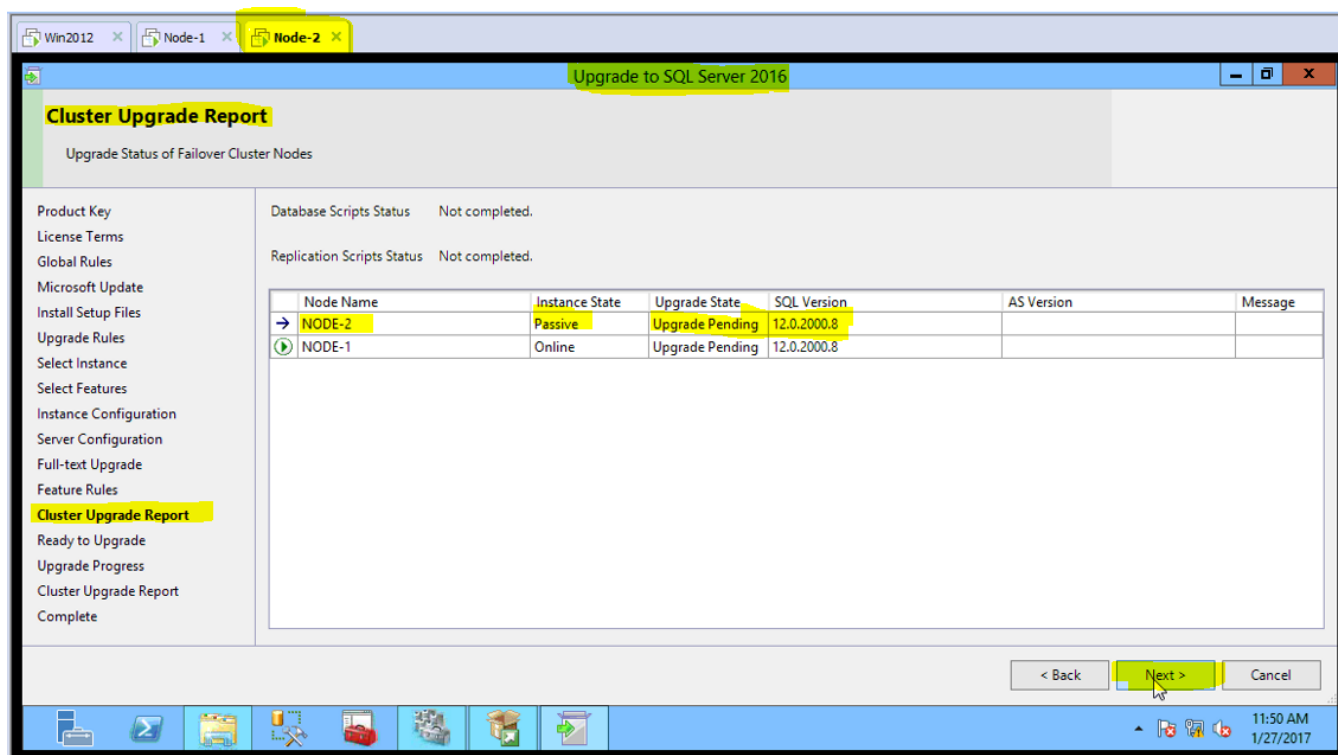
Microsoft recommends that you use a separate account for each SQL Server service.

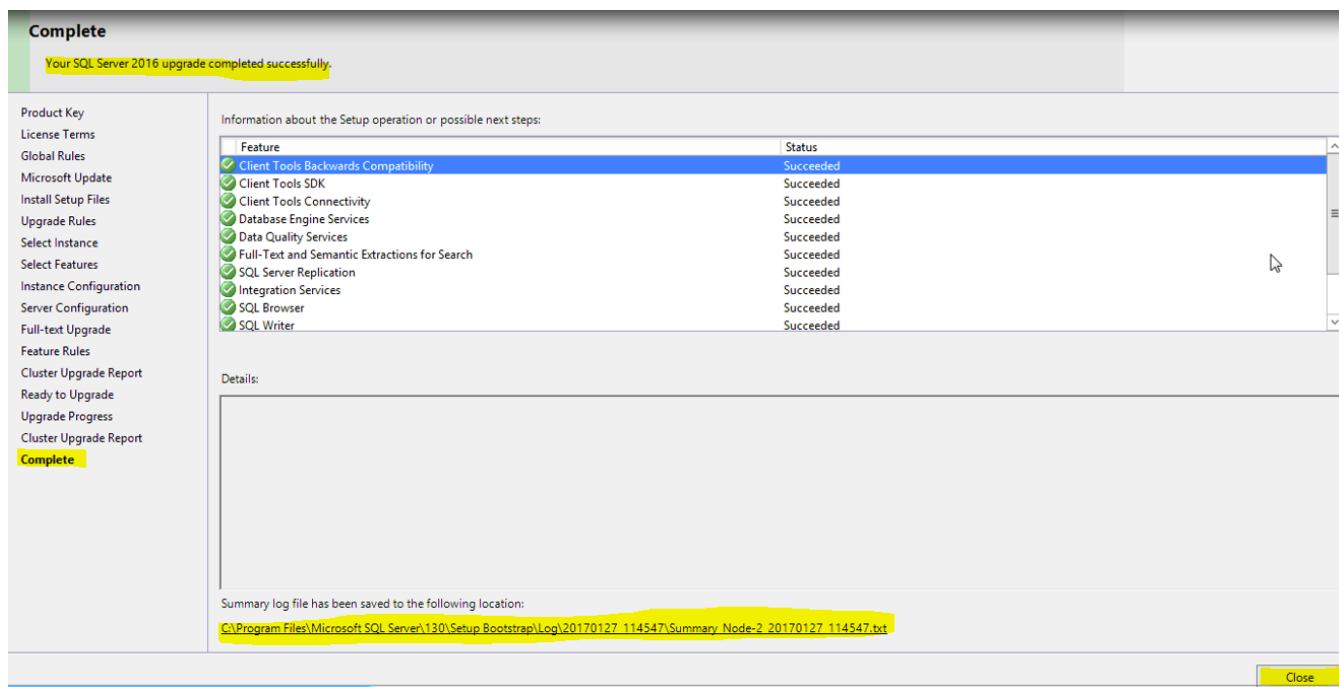
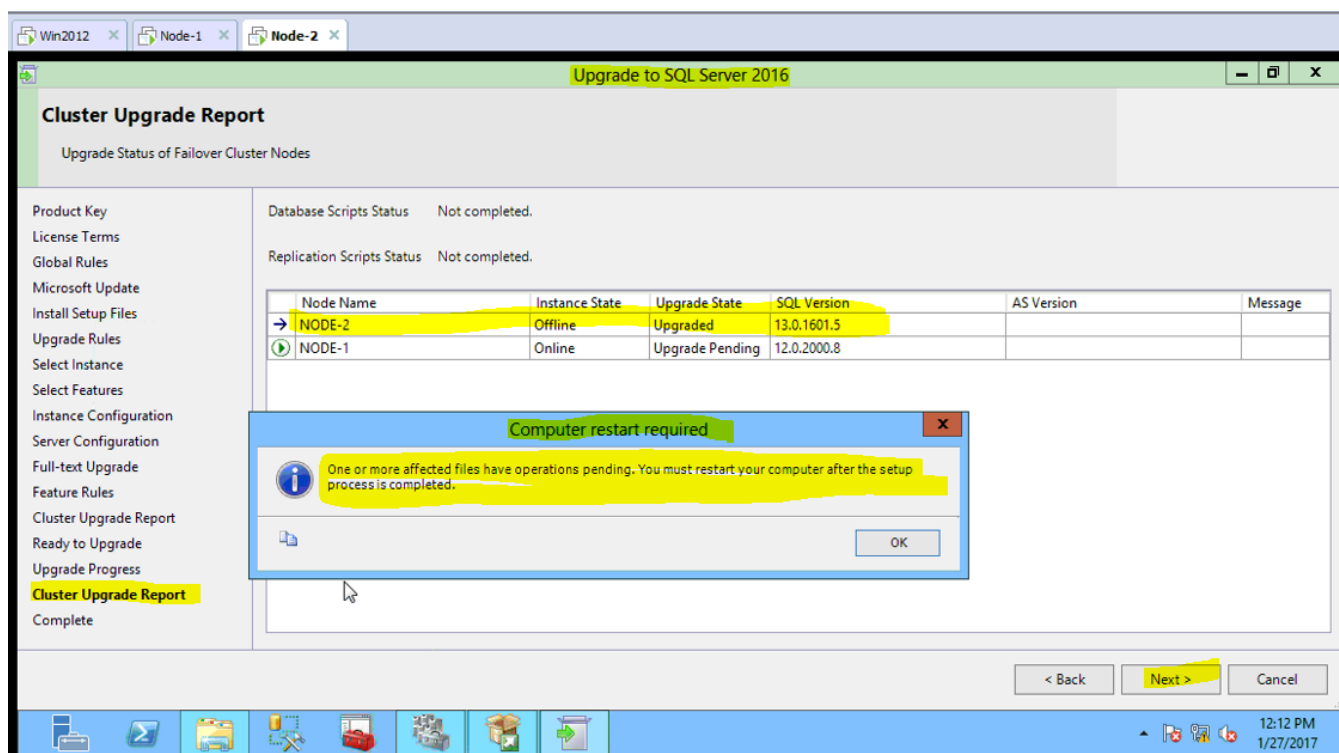
Service	Account Name	Password	Startup Type
SQL Server Integration Services 13.0	NT Service\MsDtsServer130		Automatic
SQL Full-text Filter Daemon Launcher	NT Service\MSSQLFDLauncher\$HYDERABAD		Manual

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File Edit Format View Help

Overall summary:

Final result: Passed but reboot required, see logs for details

Exit code (Decimal): 3010

Start time: 2017-01-27 11:45:47

End time: 2017-01-27 12:13:32

Requested action: Upgrade

Cluster properties:

Machine name: NODE-2

Product	Instance	Instance ID	Feature
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	Database Engine Services 1033
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	SQL Server Replication 1033
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	Full-Text and Semantic Extractions for Search 1033
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	SSMS 1033
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	Adv_SSMS 1033
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	Client Tools Connectivity 1033
SQL Server 2012	SQLDESTINATION	MSSQL11.SQLESTINATION	Client Tools Backwards Compatibility 1033

Recycle Bin

select

Microsoft Data Mi...

Upgrade Adv Report.csv

Node-2

Host Name: NODE-2

IE Version: 10.0.9200.16384

IP Address: 192.168.1.112

Logon Domain: CHAMPSDBA

Logon Server: PKSQLDBA

MAC Address: 00-0C-29-5F-3B-4C

Machine Domain: CHAMPSDBA

Memory: 2048 MB

Network Card: Intel(R) 82574L Gigabit Network Connection

Network Speed: 1 Gb/s

Network Type: Ethernet

OS Version: Windows 2012

Service Pack: No service pack

Snapshot Time: 1/11/2017 6:20 PM

Subnet Mask: 255.255.255.0

System Type: Server, Stand-alone, Terminal Server, I

User Name: clusteradmin

Volumes: B:\ 20.00 GB NTFS

Free Space: (none)

Host Name: NODE-2

IE Version: 10.0.9200.16384

IP Address: 192.168.1.112

Settings

Desktop

Control Panel

Personalization

Server info

Help

Shut down

Restart

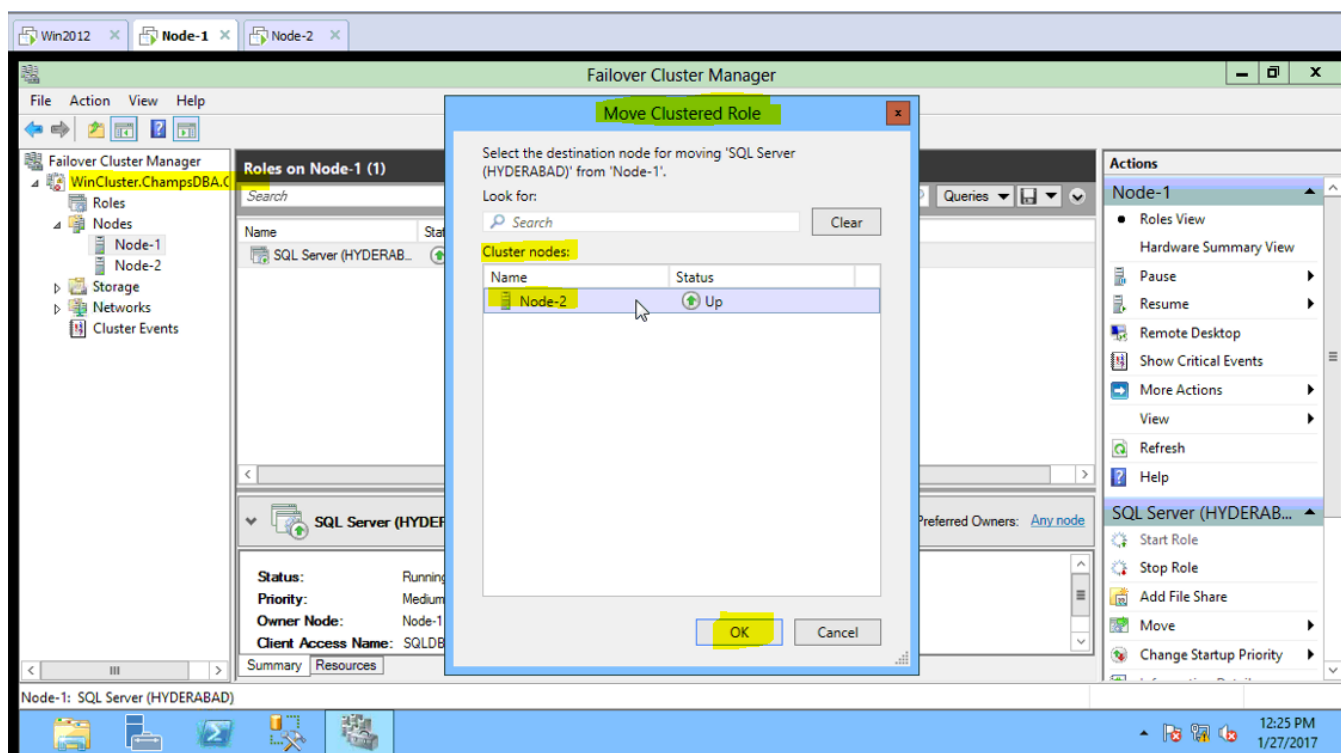
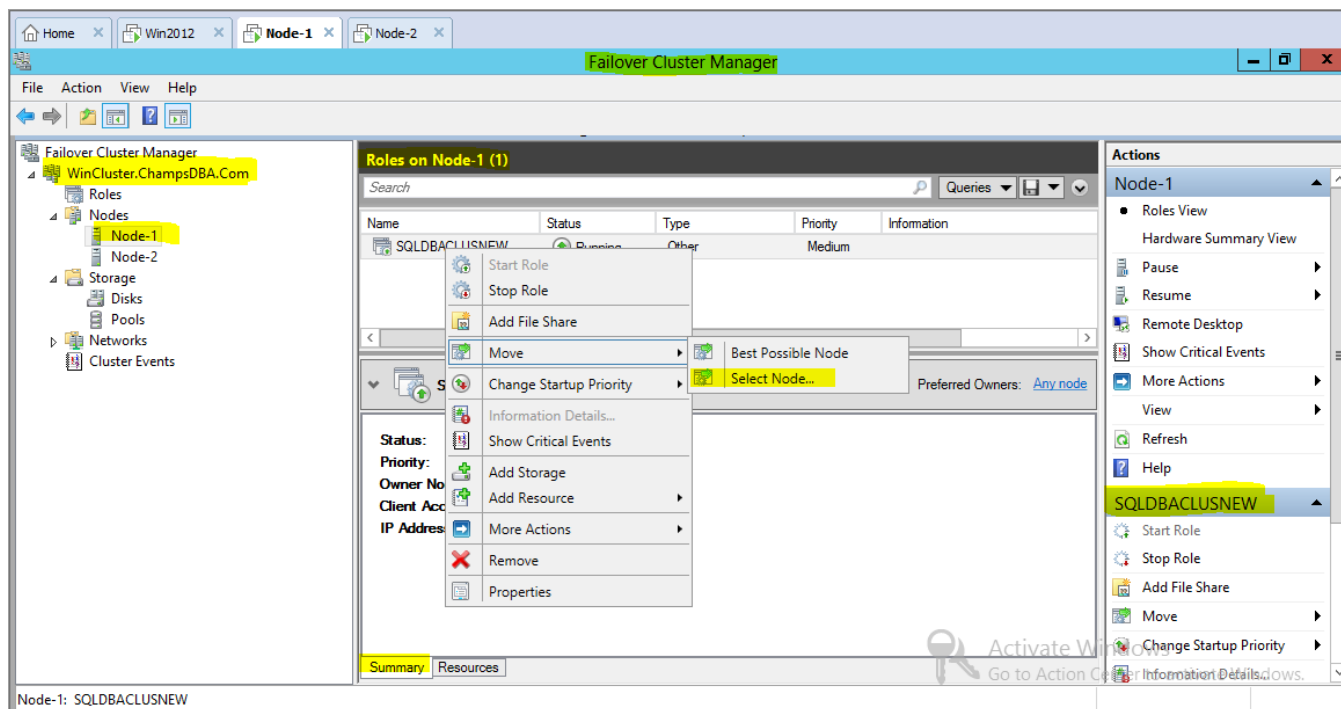
Notifications

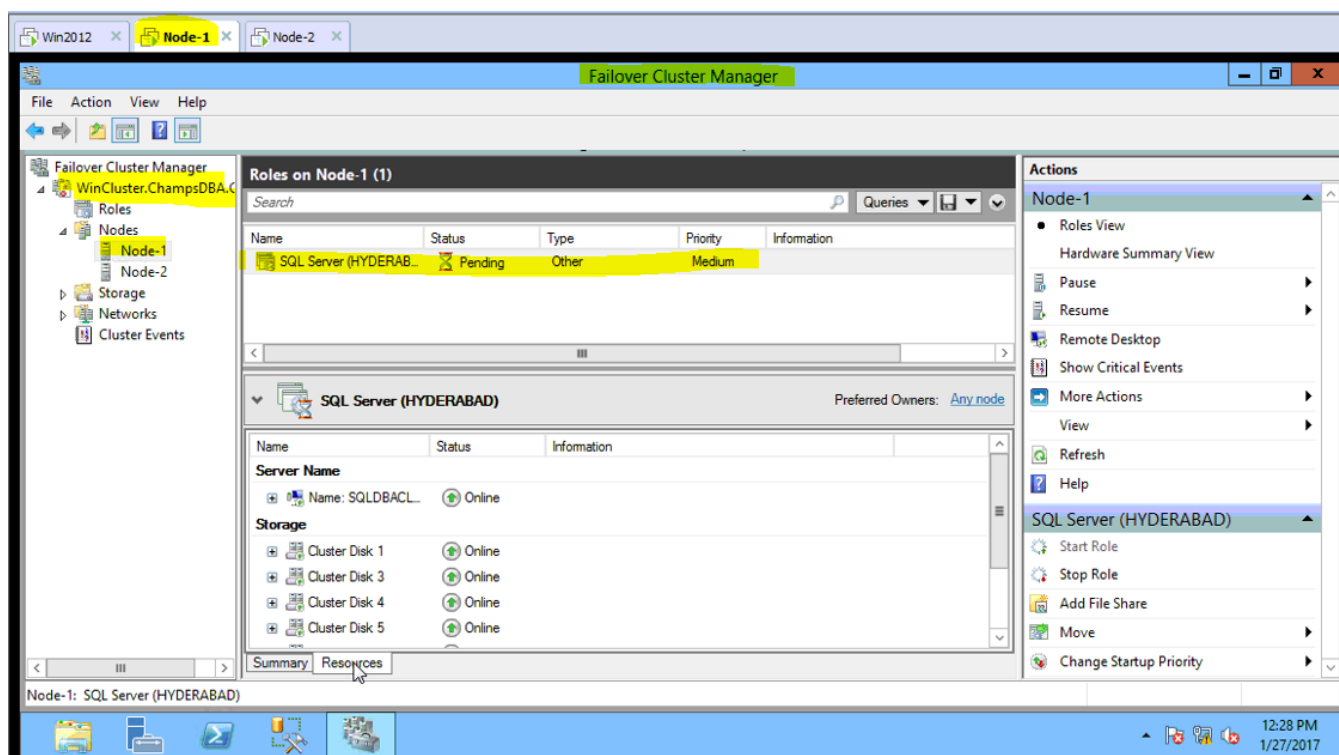
Power

Keyboard

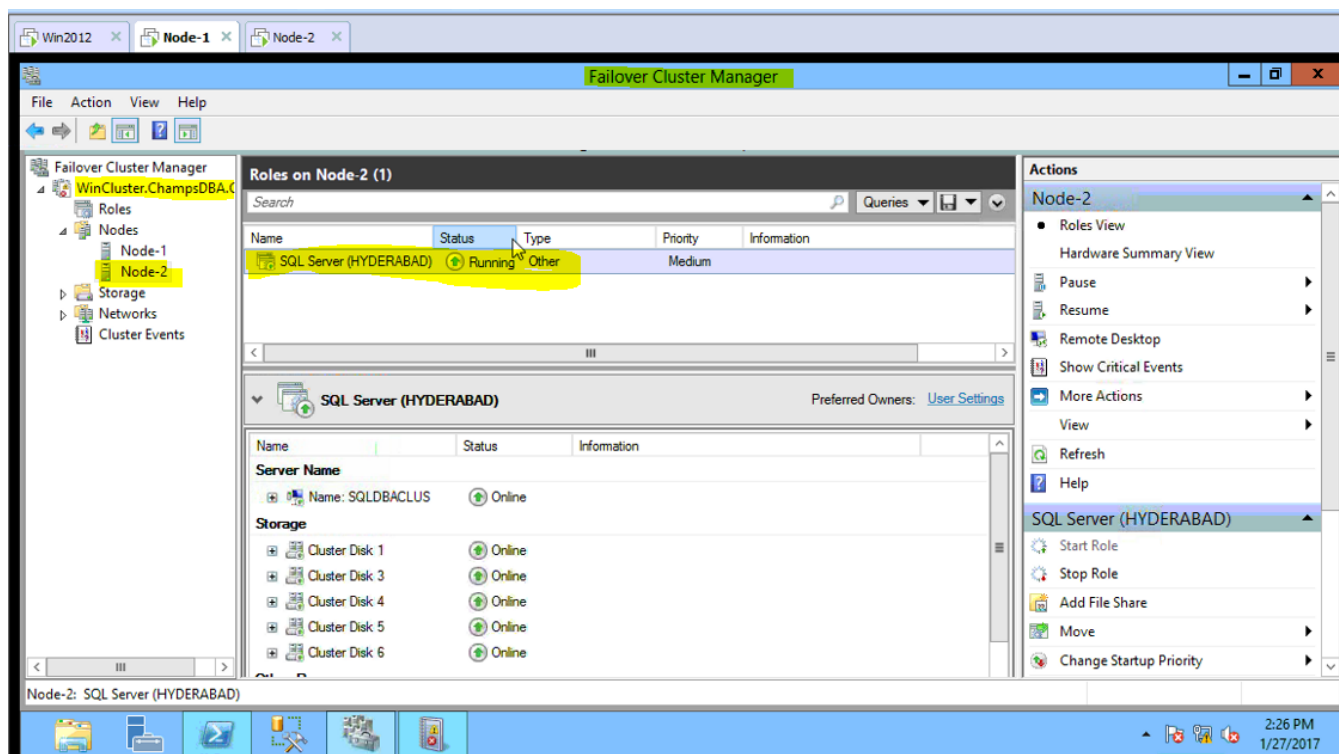
Now go to "Failover Cluster Manager" on Node-1.

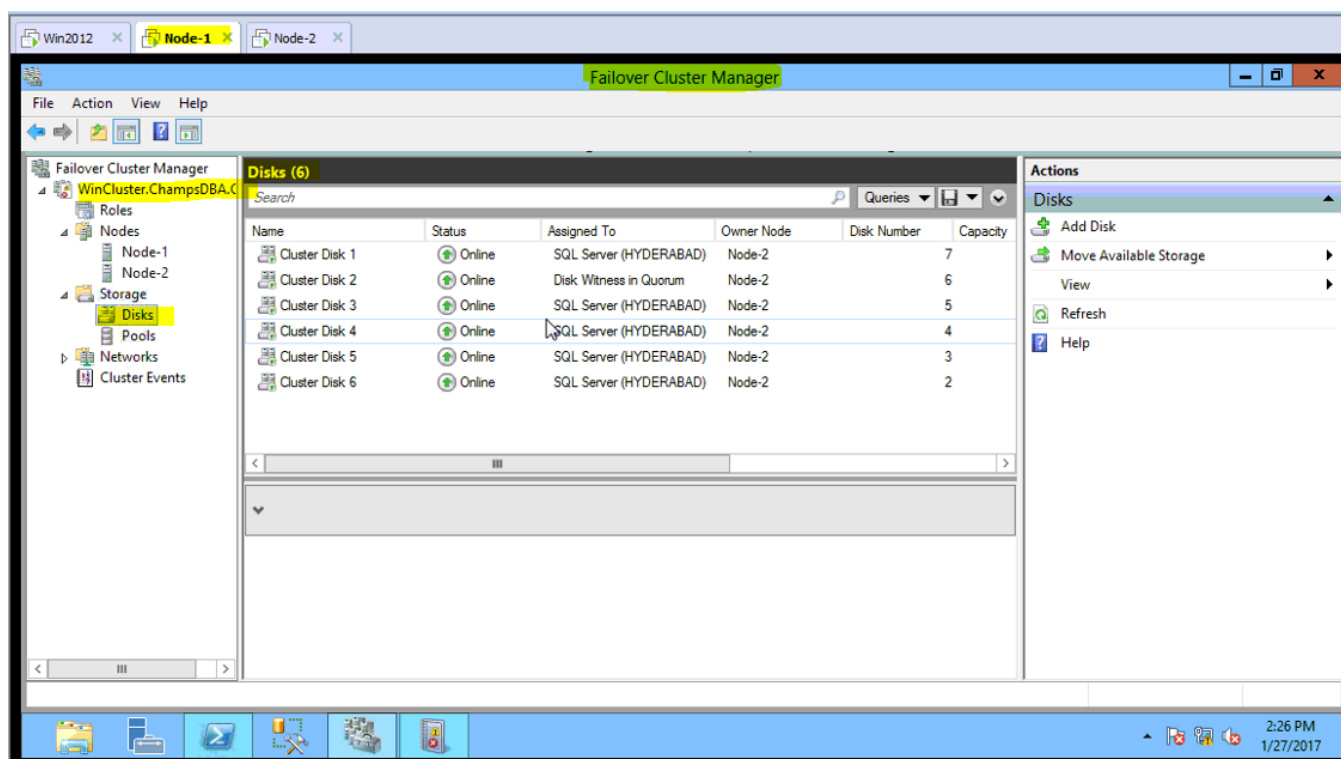
Failover the SQL Server Cluster Instance [Hyderabad] from Node-1 to Node-2 as follows.





Now SQL Server Cluster Instance [Hyderabad] is running on Node-2:

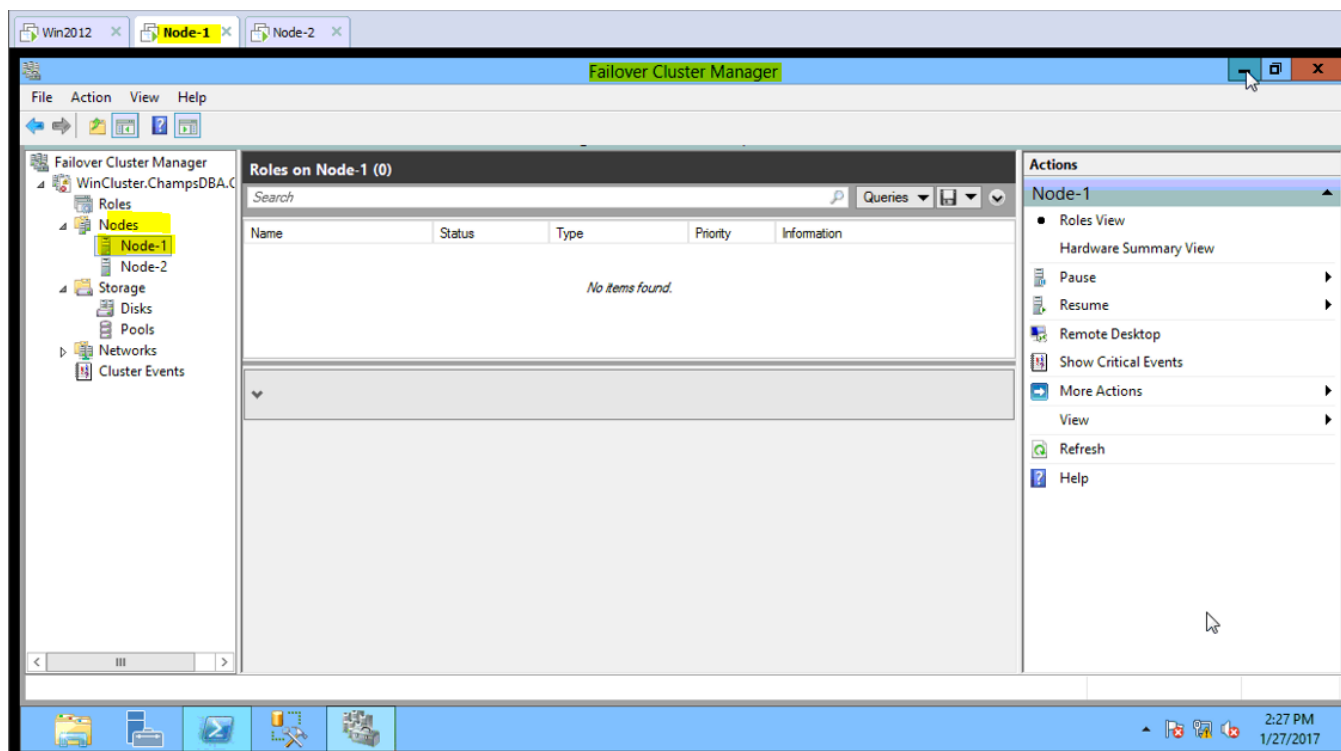




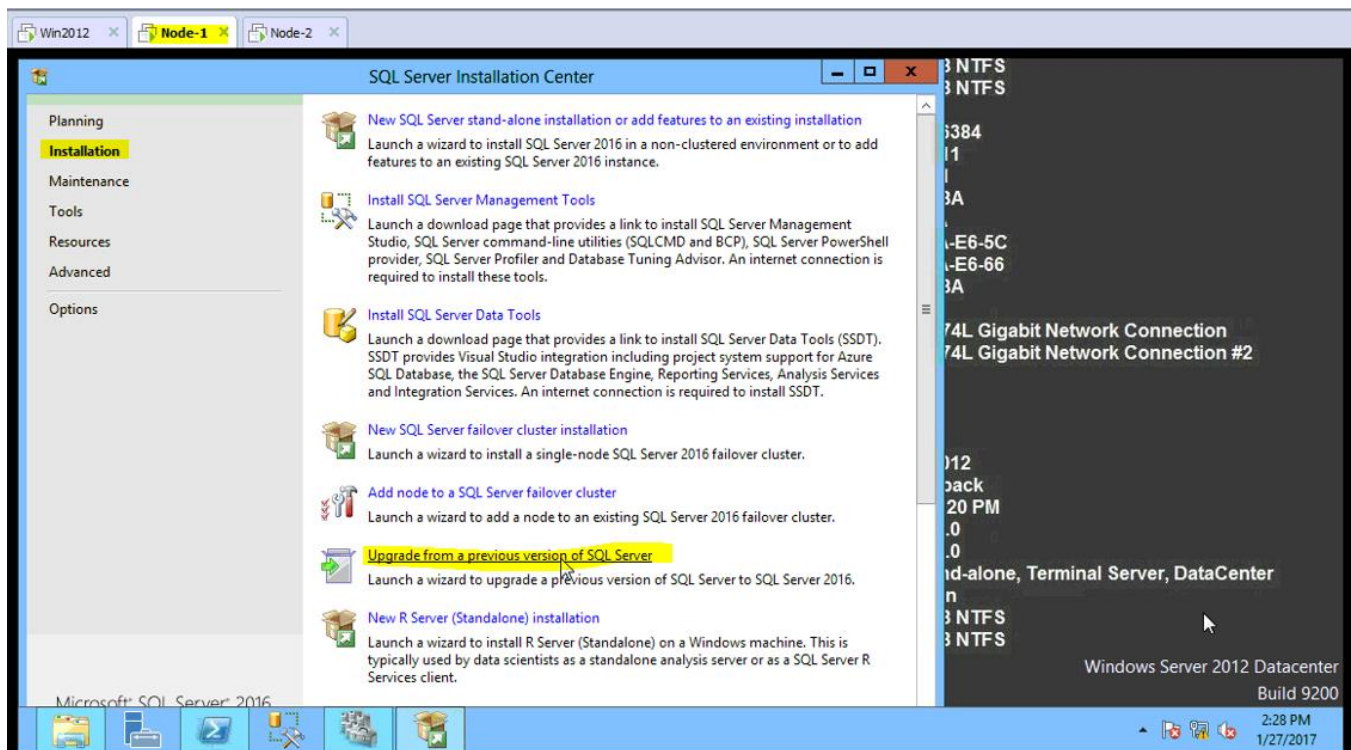
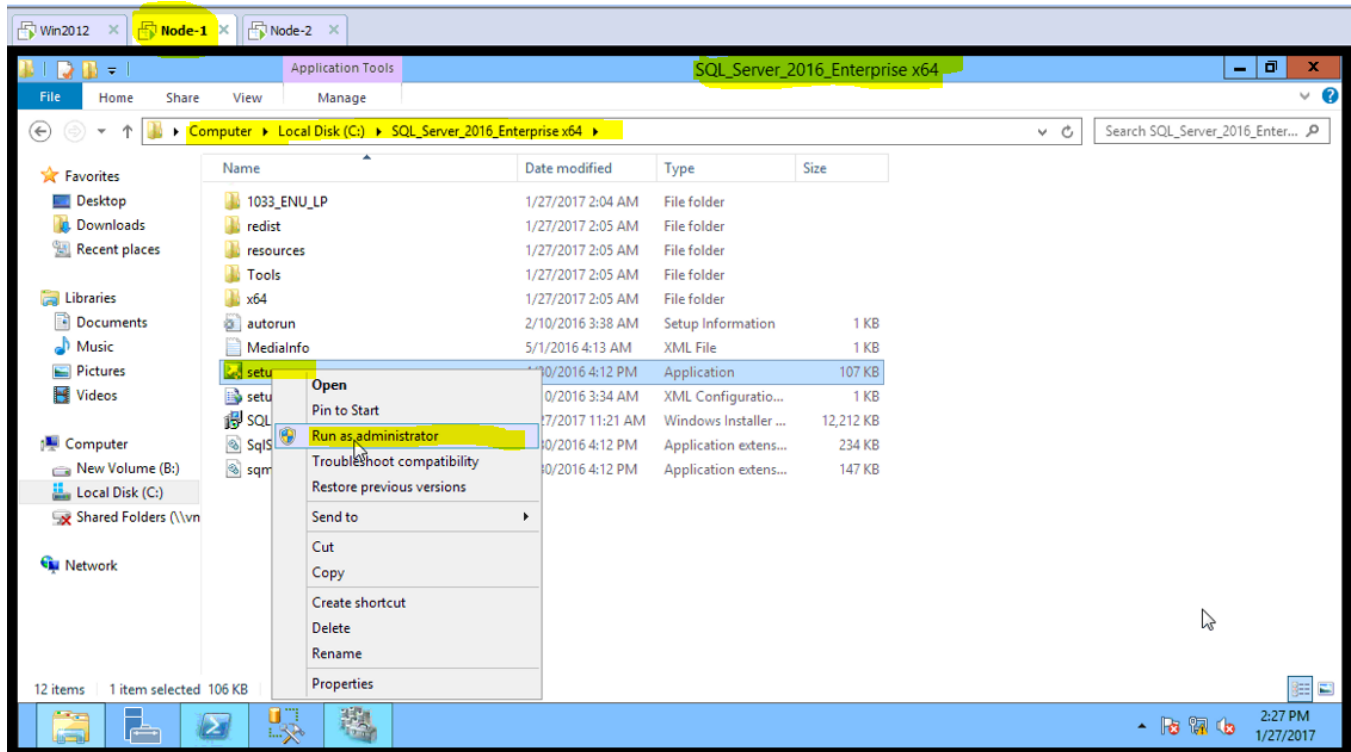
Now Upgrade the SQL Server on new "Passive" node Node-1:

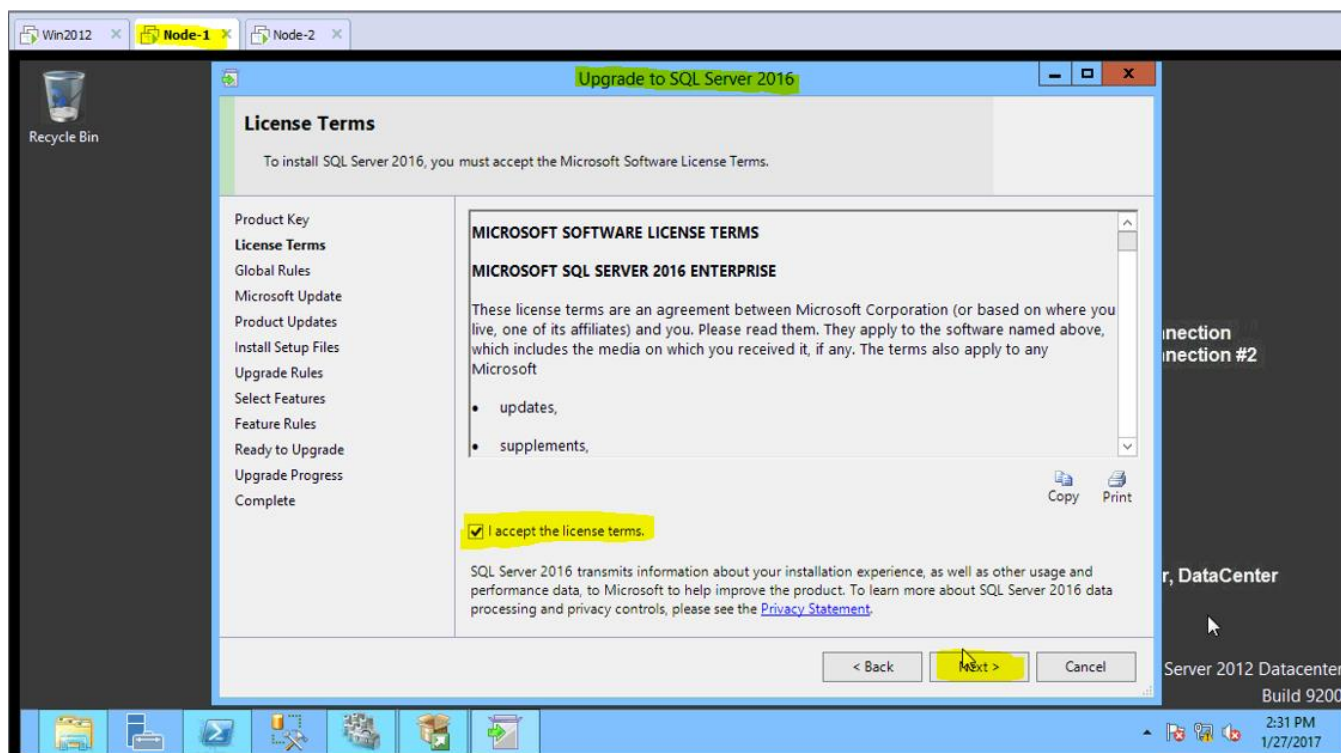
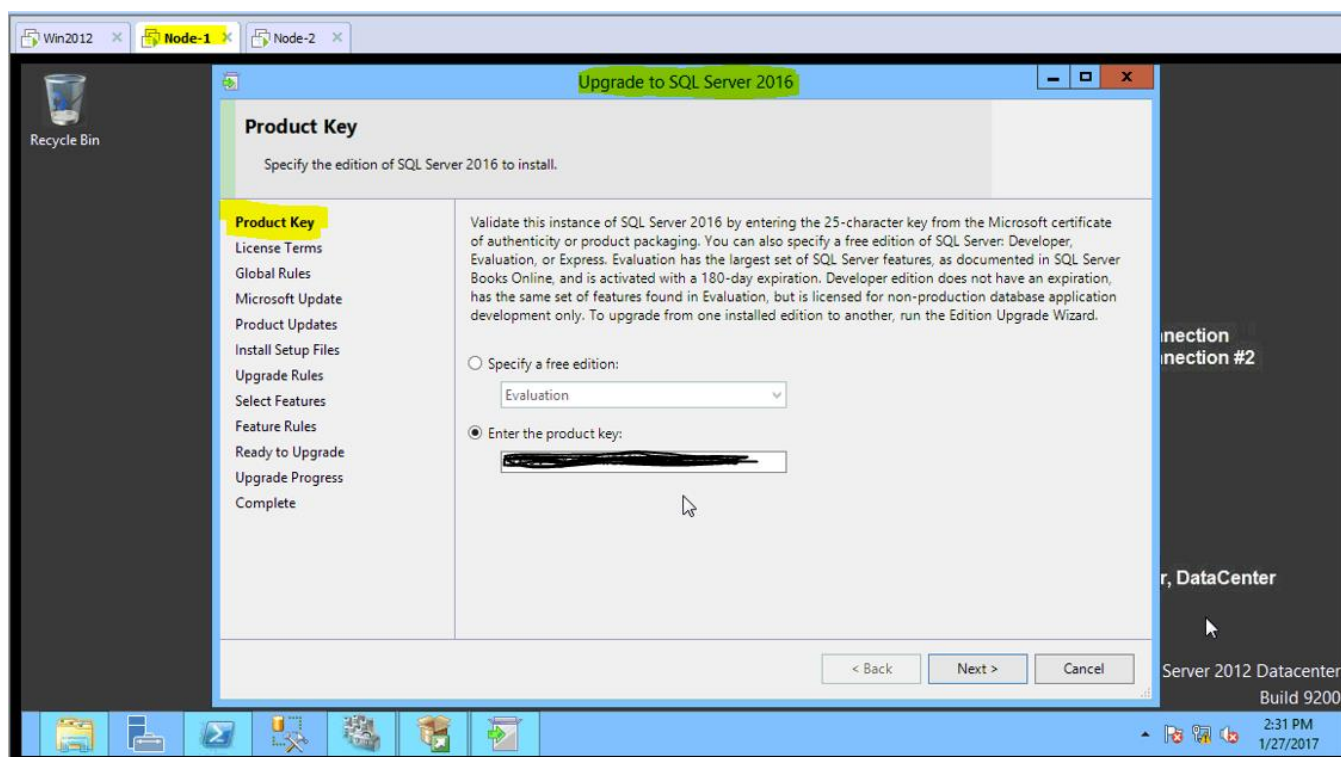
Take Remote Desktop Connection to Node-1:

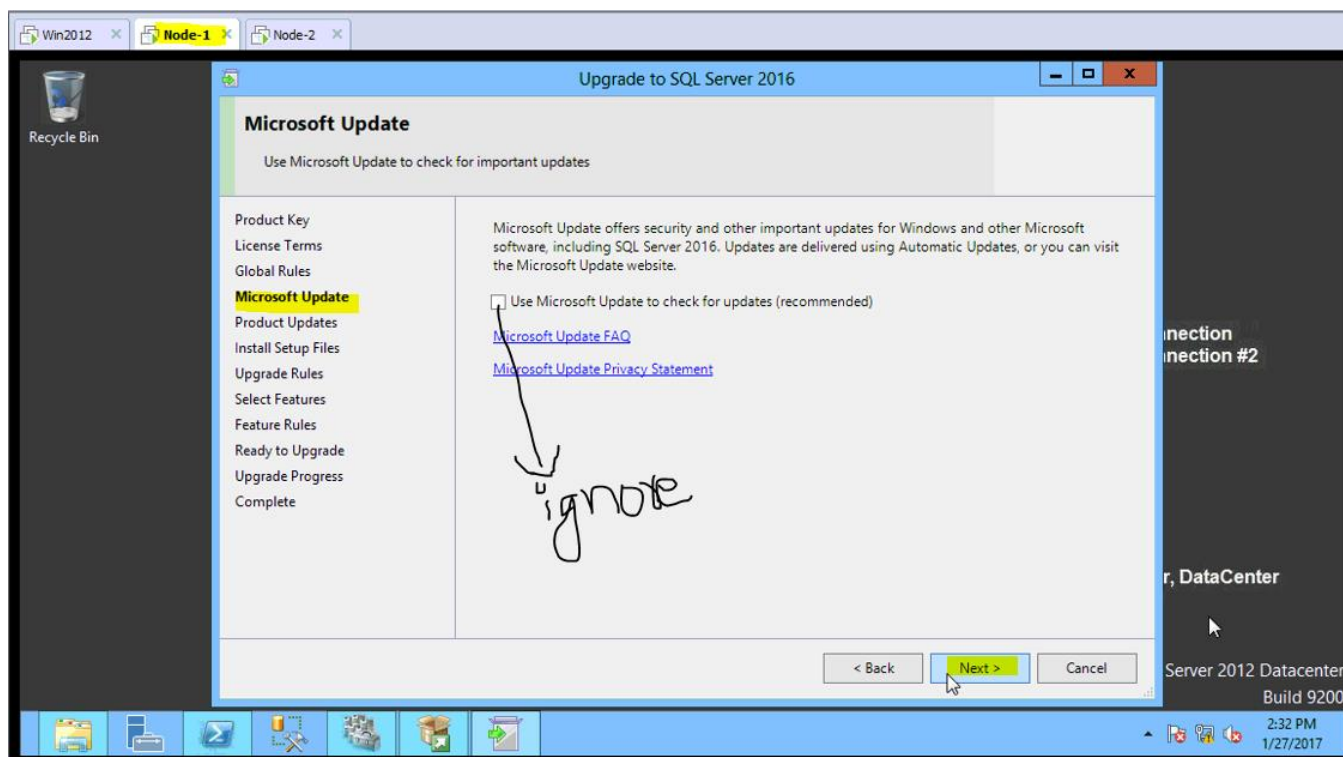
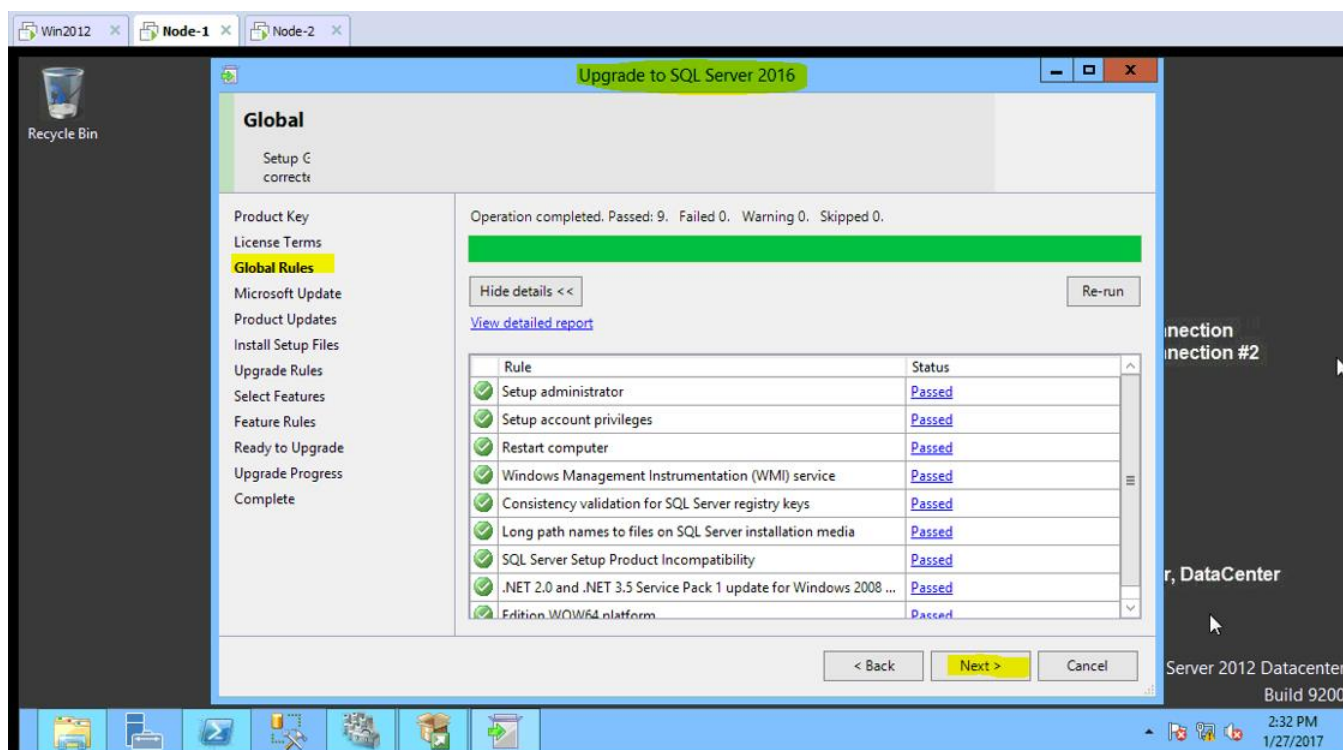
Ensure that on Node-1 own any services or disks.

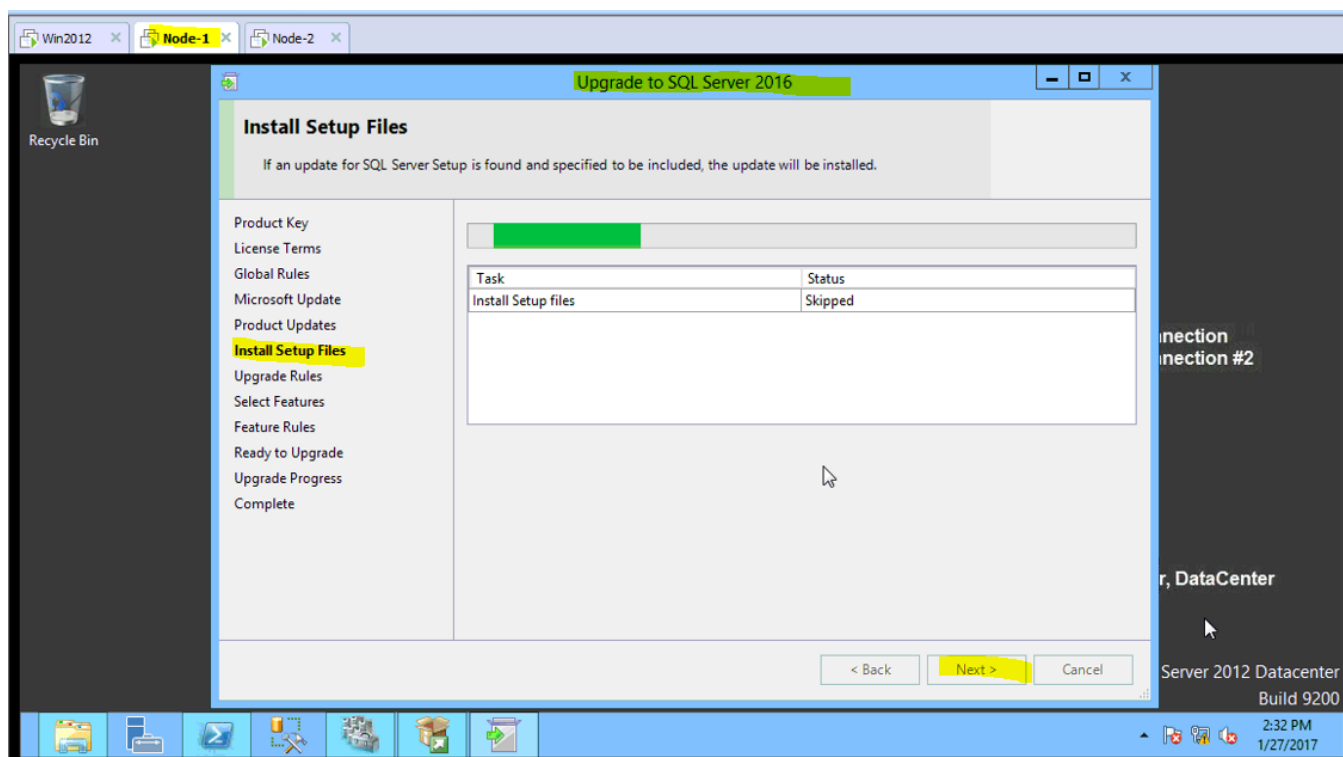
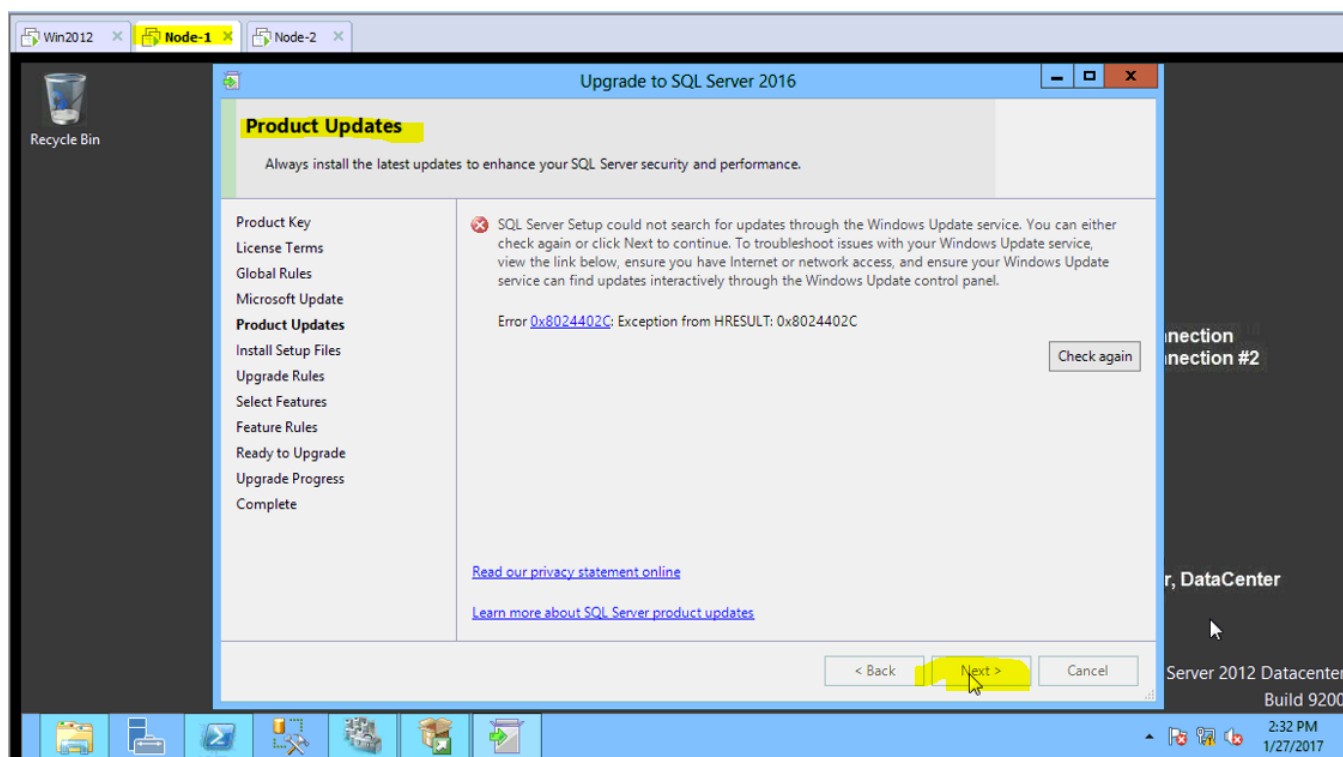


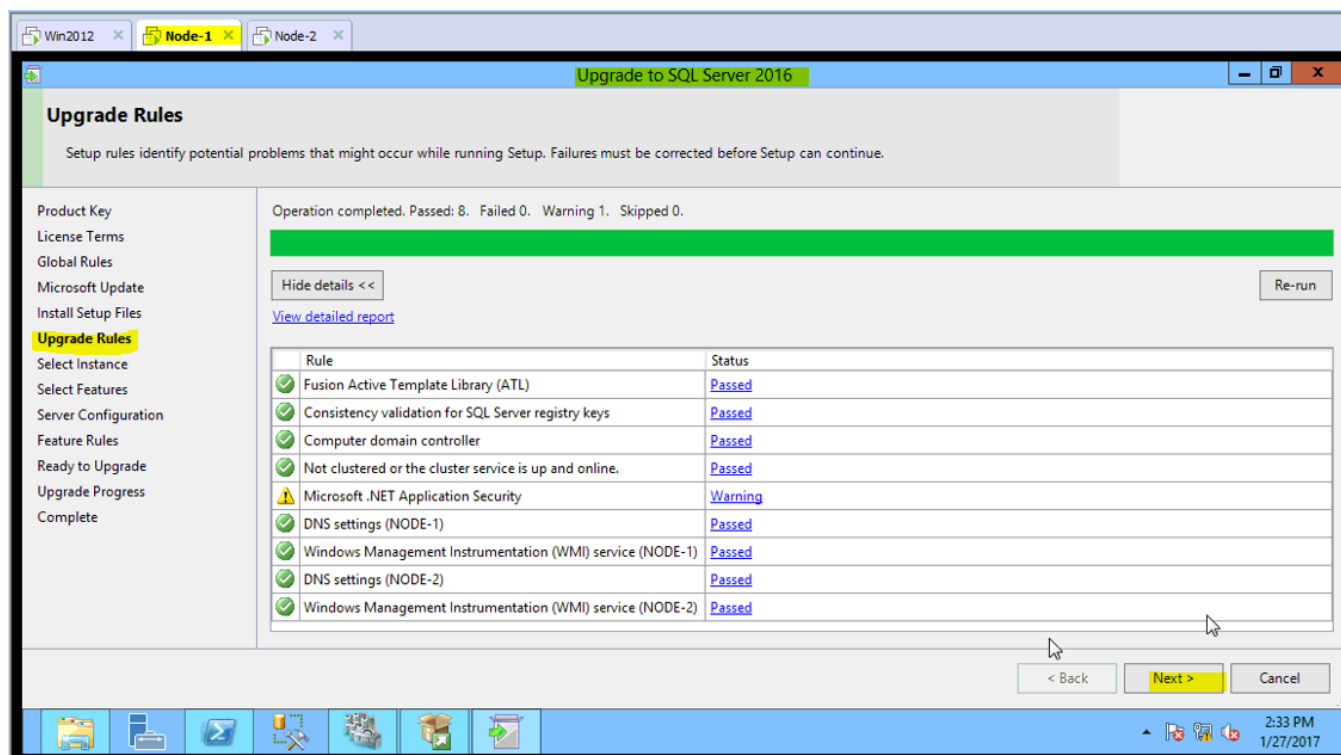
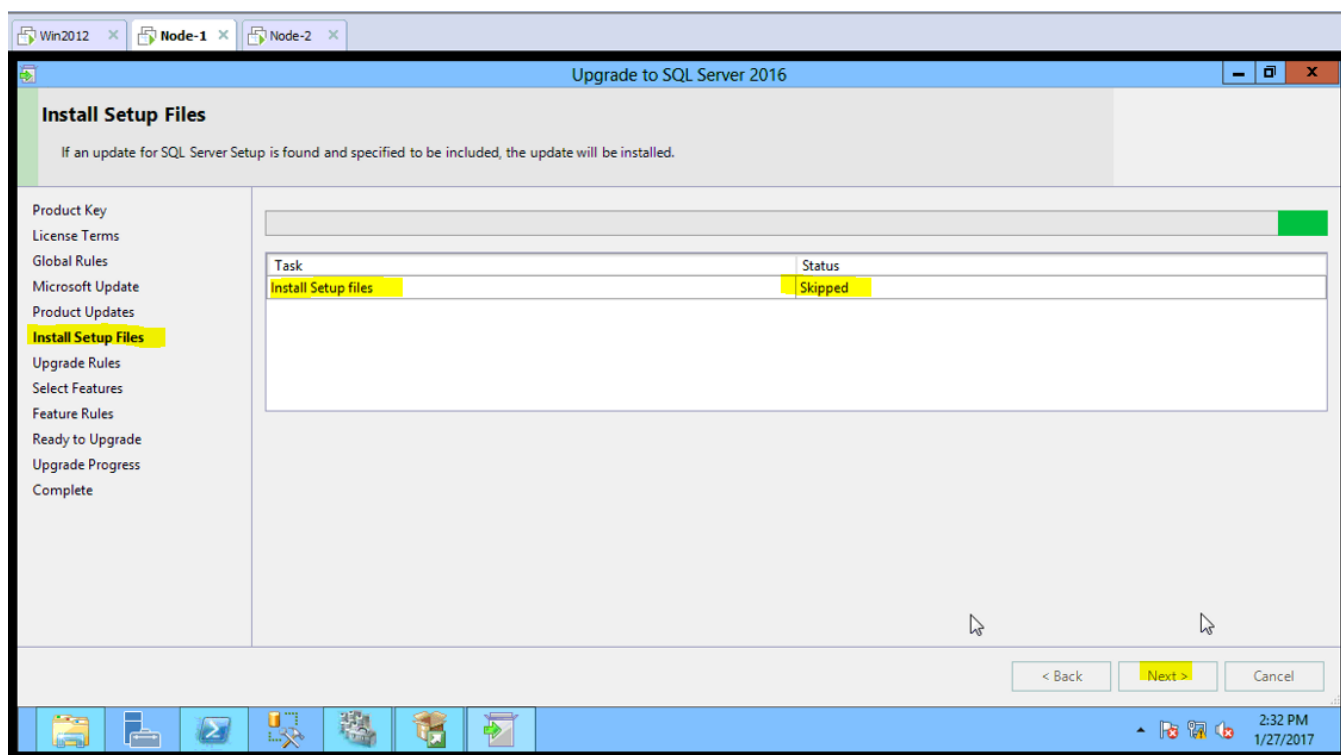
Navigate to the Path where SQL Server 2016 on Node-1:

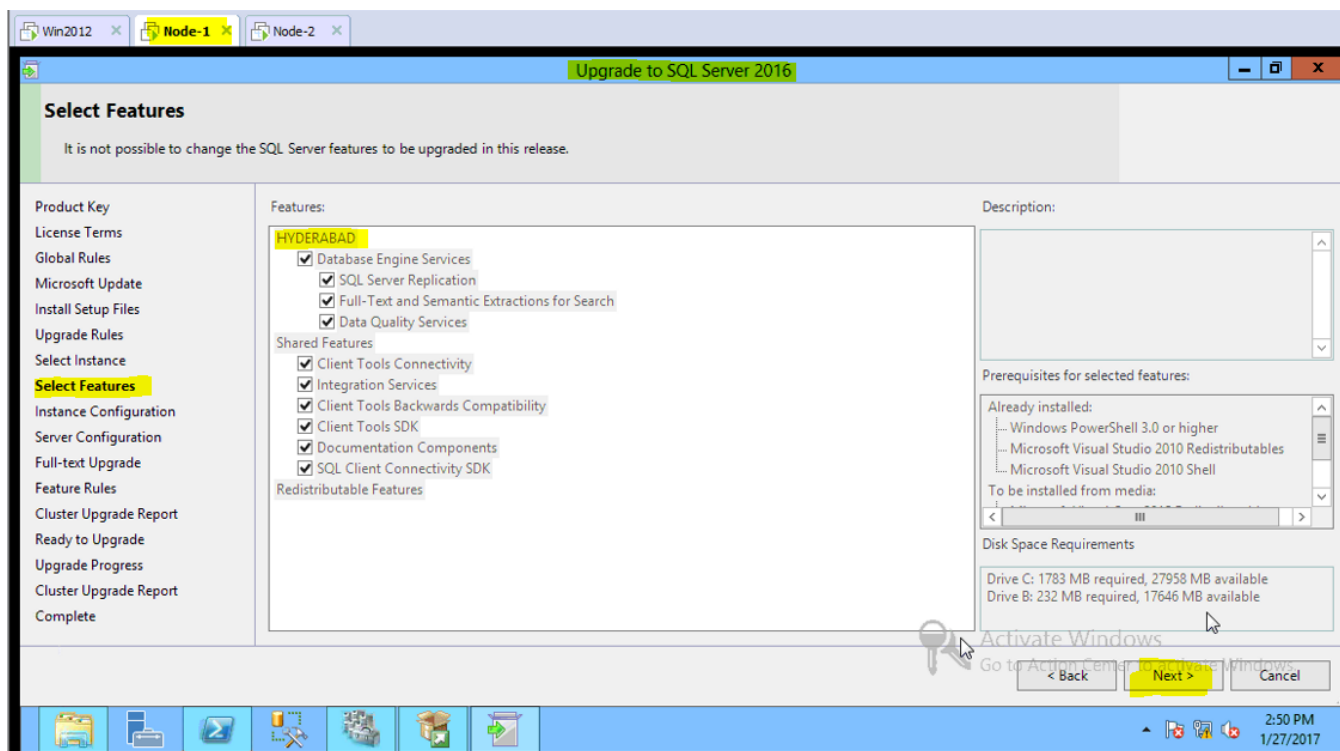
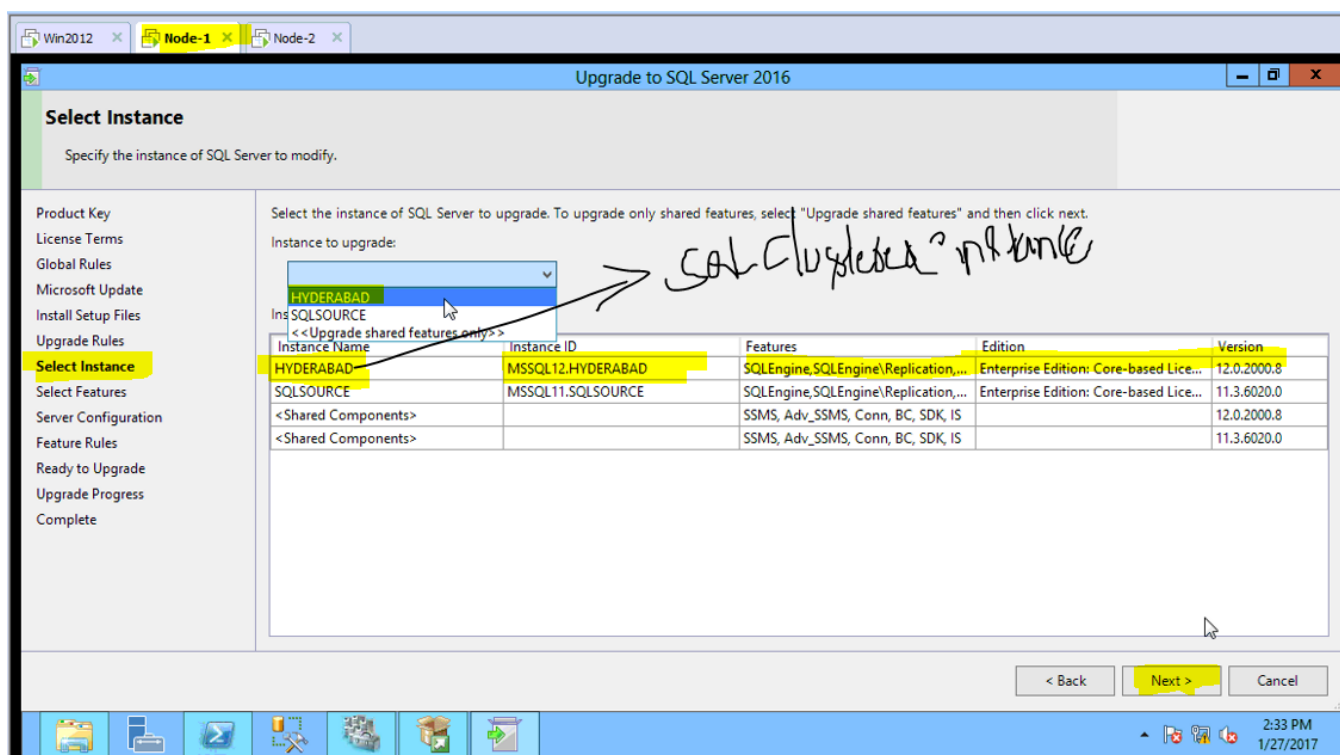












Win2012 x Node-1 x Node-2

Upgrade to SQL Server 2016

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Product Key
License Terms
Global Rules
Microsoft Update
Install Setup Files
Upgrade Rules
Select Instance
Select Features
Instance Configuration
Server Configuration
Full-text Upgrade
Feature Rules
Cluster Upgrade Report
Ready to Upgrade
Upgrade Progress
Cluster Upgrade Report
Complete

☐ Default instance
☒ Named instance: **HYDERABAD**

Instance ID: **HYDERABAD**

SQL Server directory: B:\Program Files\Microsoft SQL Server\MSSQL13.HYDERABAD

Installed instances:

Instance Name	Instance ID	Features	Edition	Version
HYDERABAD	MSSQL12.HYDERABAD	SQLEngine,SQLEngine\Replication,...	Enterprise Edition: Core-based Lice...	12.0.2000.8
SQLSOURCE	MSSQL11.SQLSOURCE	SQLEngine,SQLEngine\Replication,...	Enterprise Edition: Core-based Lice...	11.3.6020.0
<Shared Components>		SSMS, Adv_SSMS, Conn, BC, SDK, IS		12.0.2000.8
<Shared Components>		SSMS, Adv_SSMS, Conn, BC, SDK, IS		11.3.6020.0

Activate Windows
Go to Action Center to activate Windows.

< Back Next > Cancel

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1/27/2017

Win2012 x Node-1 x Node-2

Upgrade to SQL Server 2016

Server Configuration

Specify the service accounts and collation configuration.

Product Key
License Terms
Global Rules
Microsoft Update
Install Setup Files
Upgrade Rules
Select Instance
Select Features
Instance Configuration
Server Configuration
Full-text Upgrade
Feature Rules
Cluster Upgrade Report
Ready to Upgrade
Upgrade Progress
Cluster Upgrade Report
Complete

Service Accounts

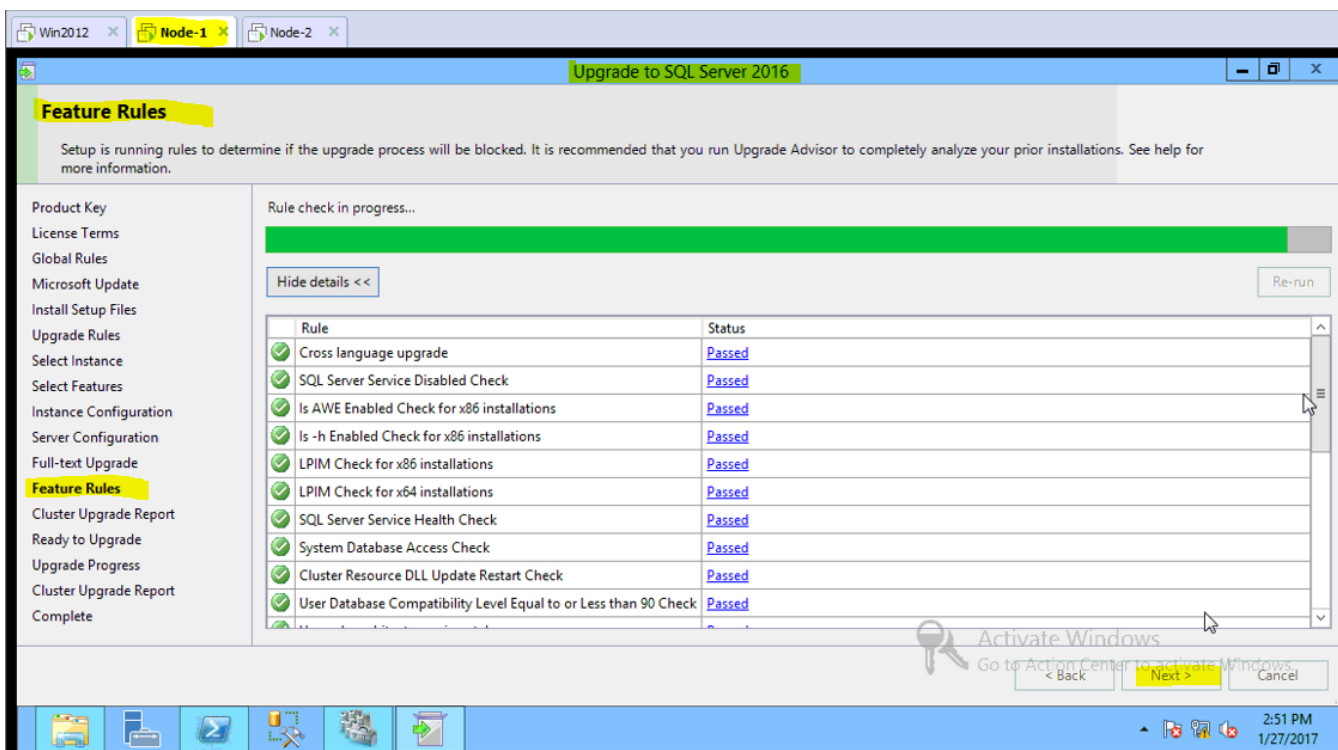
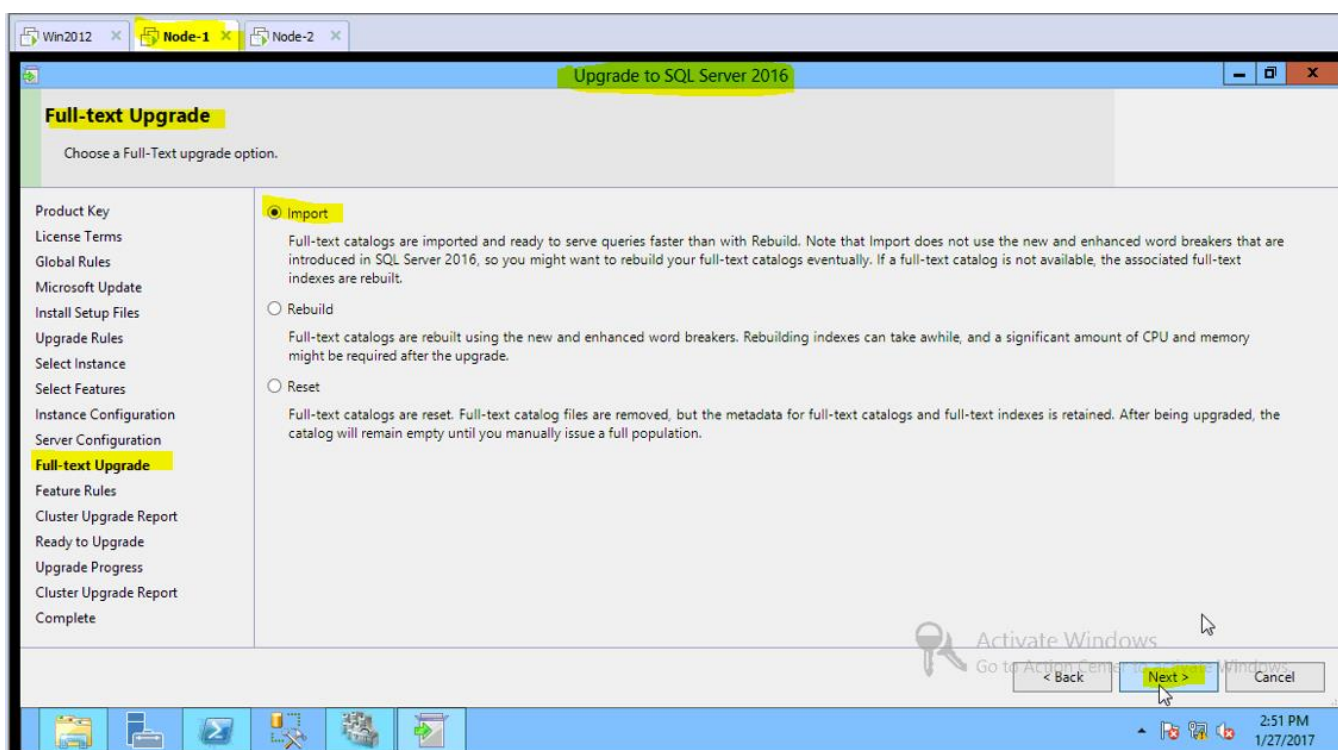
Microsoft recommends that you use a separate account for each SQL Server service.

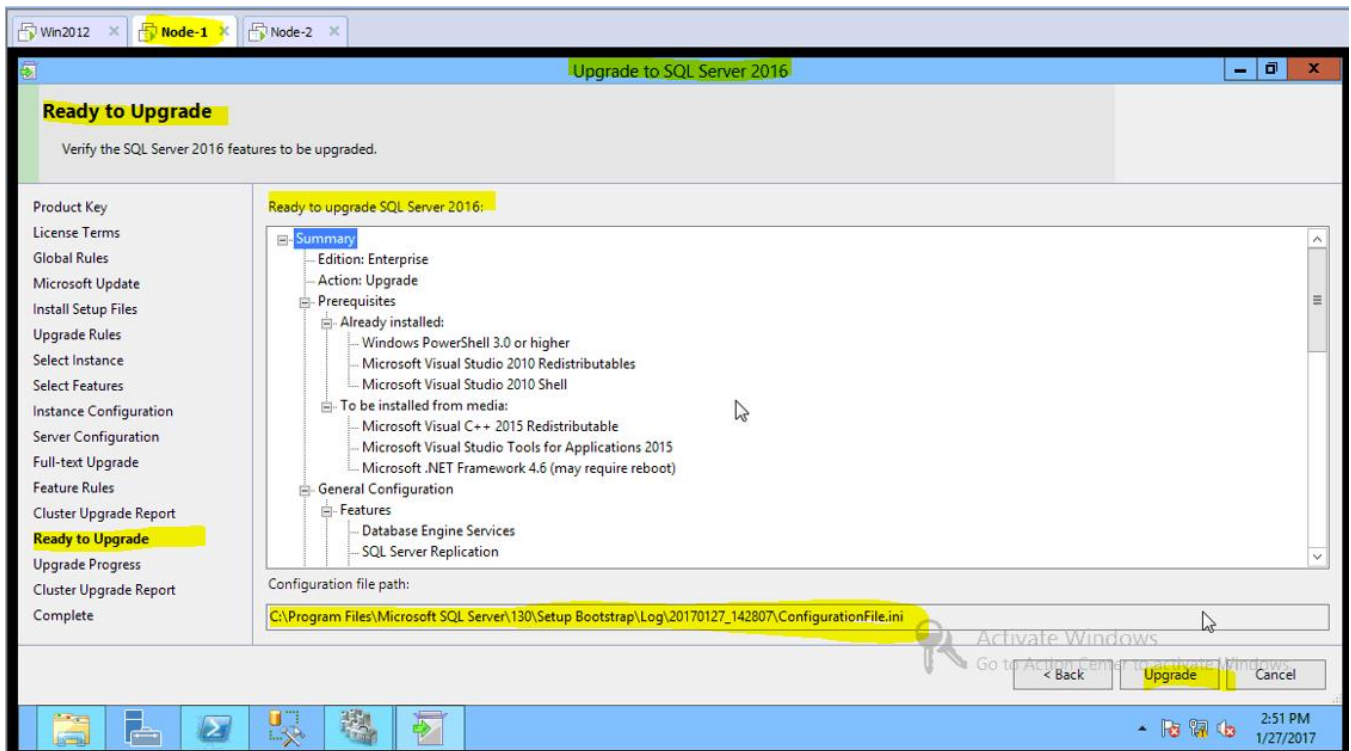
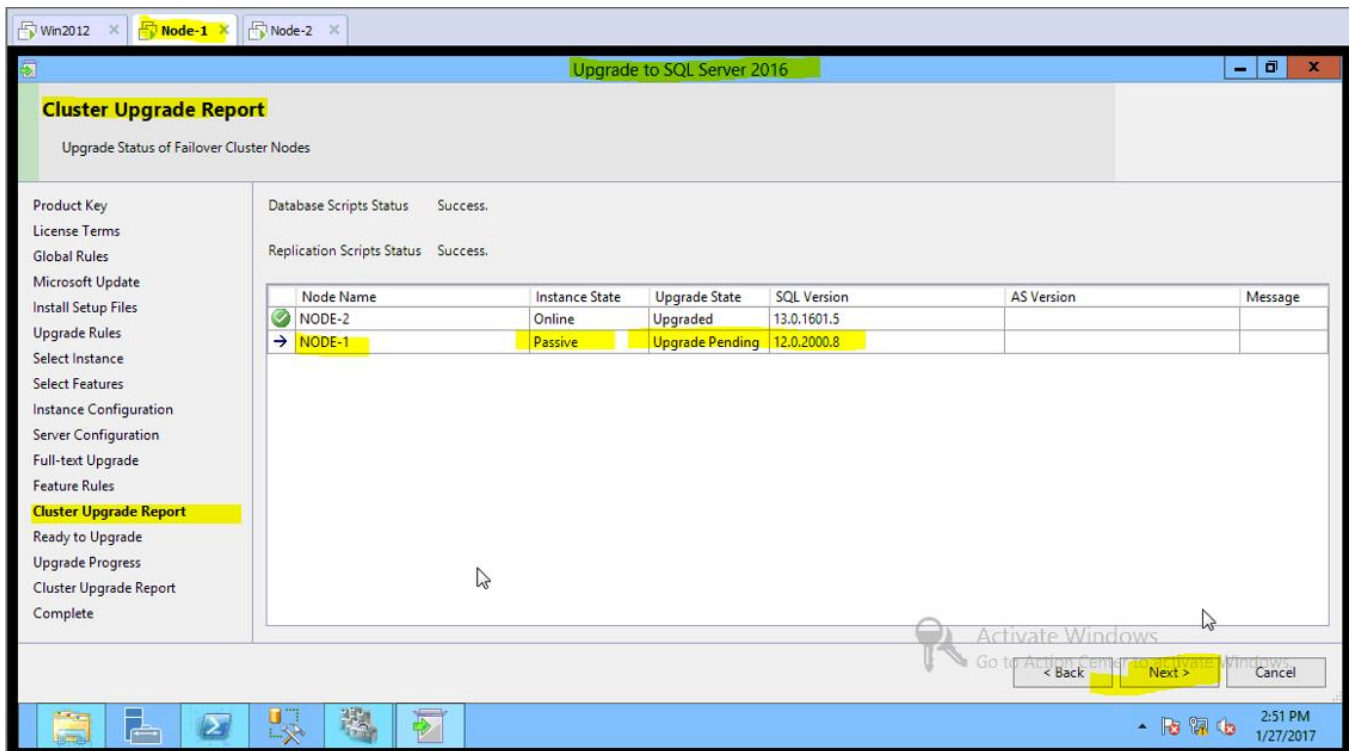
Service	Account Name	Password	Startup Type
SQL Server Integration Services 13.0	NT Service\MsDtsServer130		Automatic
SQL Full-text Filter Daemon Launcher	NT Service\MSSQLFDLauncher\$HYDERABAD		Manual

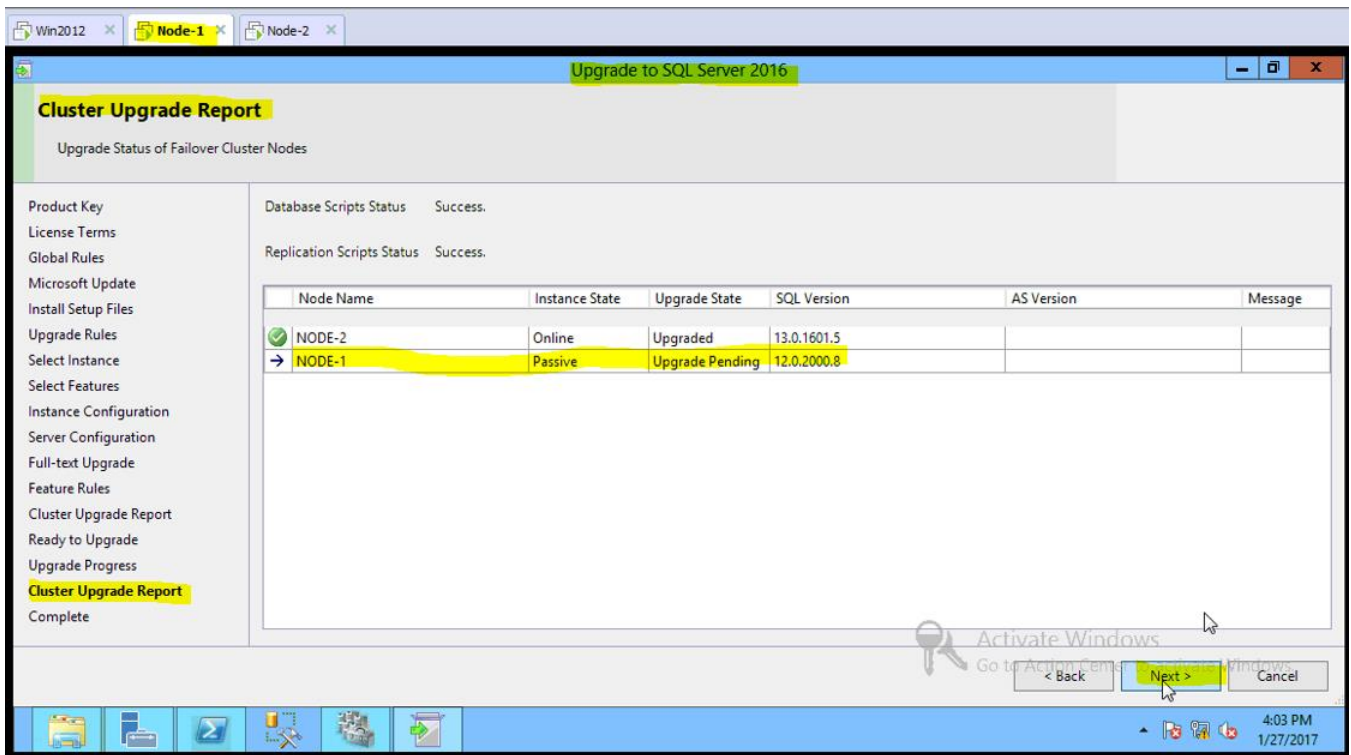
Activate Windows
Go to Action Center to activate Windows.

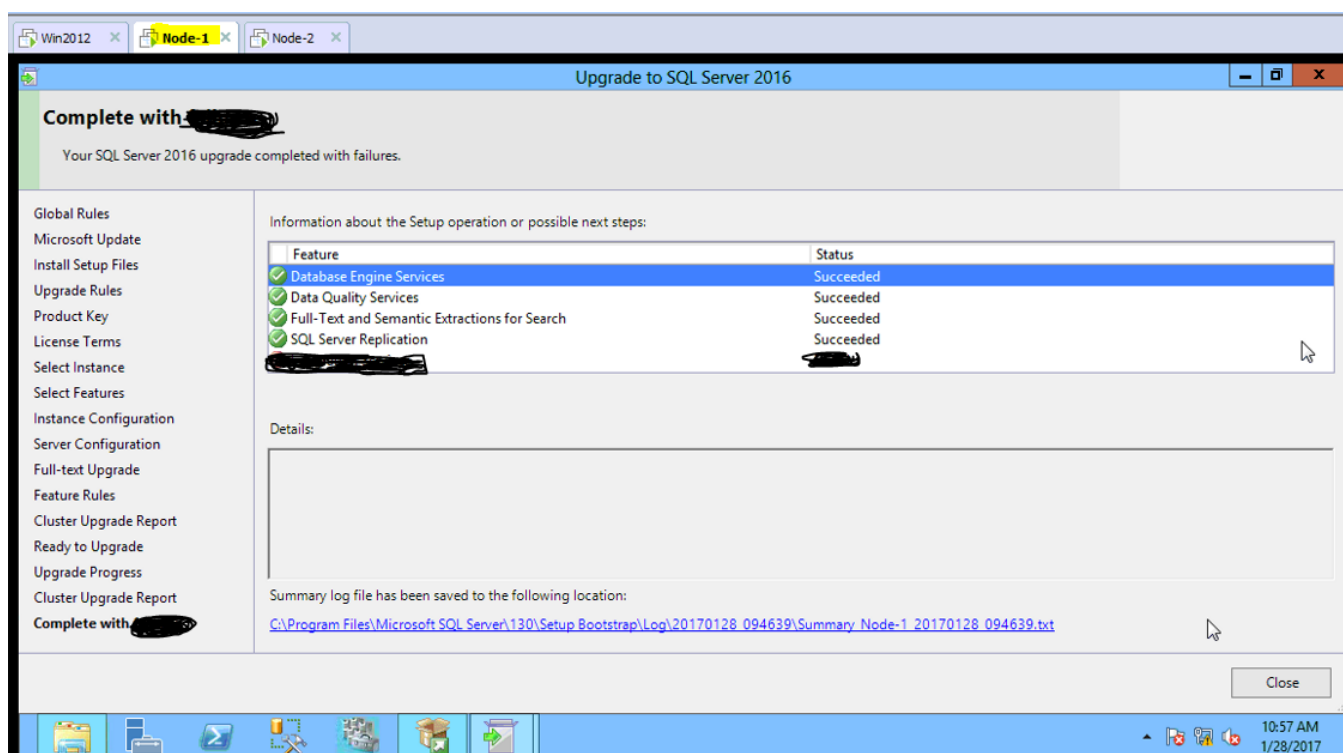
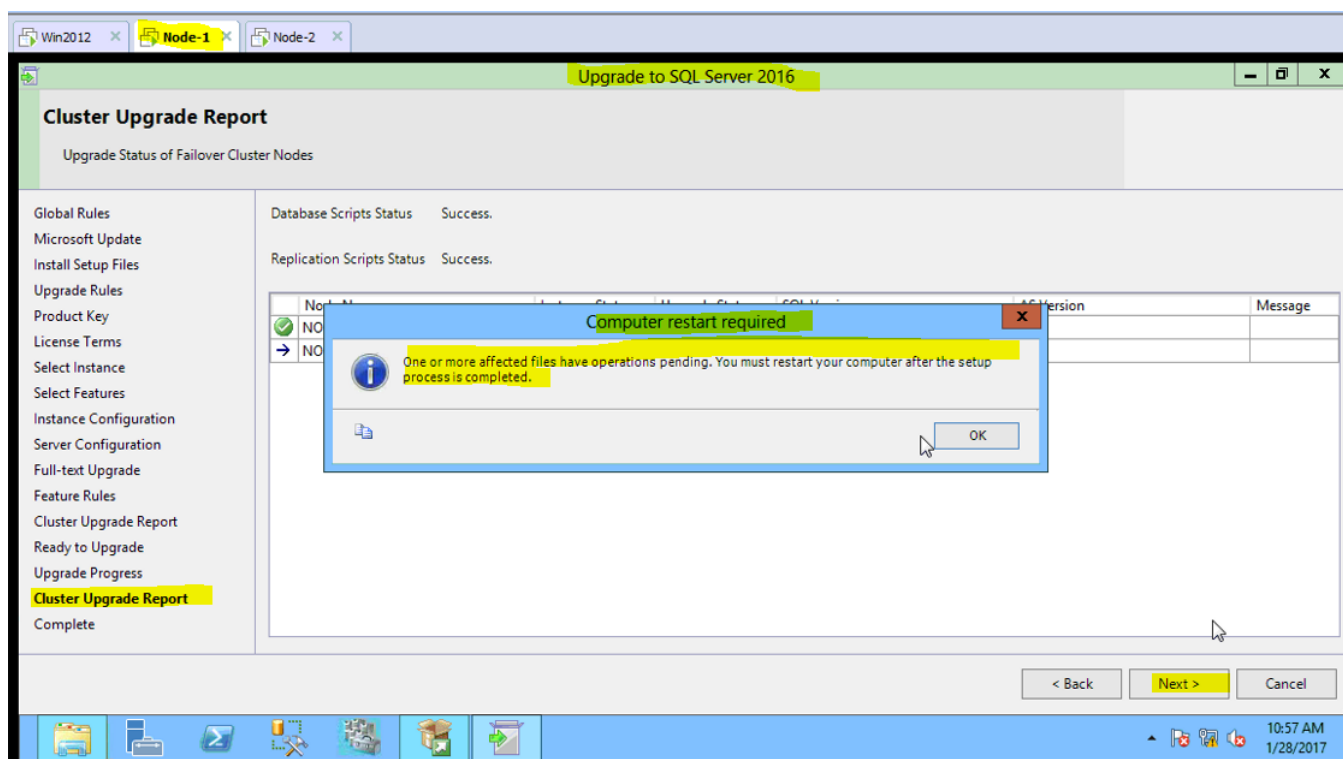
< Back Next > Cancel

2:50 PM
1/27/2017







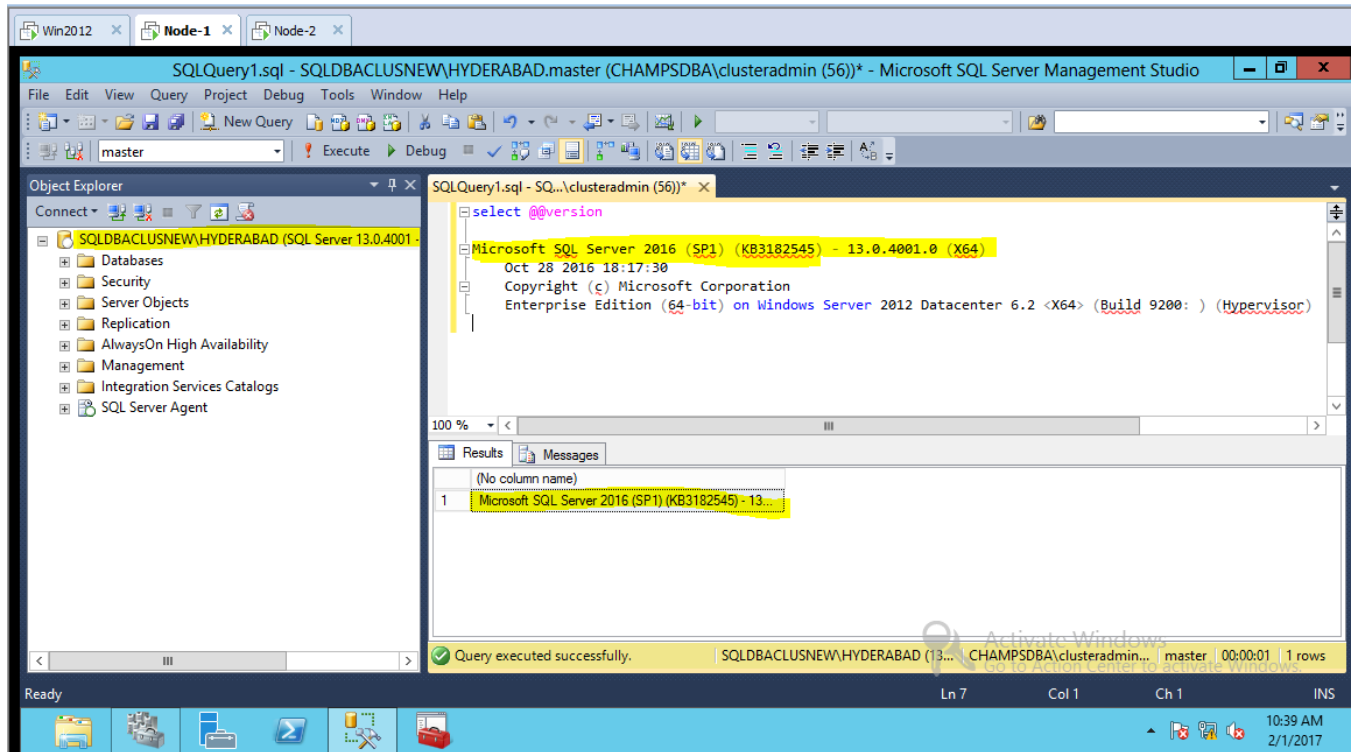


Reboot the Passive Node-1.

Failback the services from Node -2 to Node-1.

After Up gradation to SQL Server 2016:

Select @@Version



Microsoft SQL Server 2016 (SP1) (KB3182545) - 13.0.4001.0 (X64)

Oct 28 2016 18:17:30

Copyright (c) Microsoft Corporation

Enterprise Edition (64-bit) on Windows Server 2012 Datacenter 6.2 <X64> (Build 9200:)
(Hypervisor)

Note: Here after up gradation to SQL Server 2016, I have applied SP1.